Quantitative Hamiltonian path counts and degree stability

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Let $H(T)$ count directed Hamiltonian paths in an n-vertex tournament, and put $P(n) = \max_{|V(T)|=n} H(T)$, $\mu_n = n!/2^{n-1}$ and $L = \cosh 1/\cos 1$. We prove

$$P(n) = \big(L + O(n^{-1})\big)\mu_n.$$

This refines the sharp leading asymptotic of Divoux and Wang to a uniform $O(n^{-1})$ error. Retaining the degree-imbalance penalty through nonprincipal scaling and the path convolution also proves degree stability: uniformly for $0 \le \varepsilon \le (L-1)/2$ and sufficiently large n, if $H(T)/\mu_n \ge L - \varepsilon$, then $\sum_v (d^+(v) - (n-1)/2)^2 = O(\varepsilon n^2 + n)$. Exact maximizers therefore have bounded mean square degree deviation. The permanent approximation is used as an analytic tool with an independent quantitative proof, not as a new determinant main term. As related applications of the same spectral ratio, we derive Wormald's original angular Hessian independently of Bhattacharjee's overlapping result and obtain fixed-base expectation extrema and strict monotonicity of the regular-block constants.

1. Introduction

A tournament is an orientation of a complete graph. Write $H(T)$ for the number of directed Hamiltonian paths in a tournament T, and $C(T)$ for the number of directed Hamiltonian cycles. Paths are counted as vertex orderings, whereas cyclic starting points are identified for cycles. Let

$$P(n) = \max_{|V(T)|=n} H(T), \qquad \mu_n = \frac{n!}{2^{n-1}}, \qquad \nu_n = \frac{(n-1)!}{2^n}.$$

In a uniform random tournament, each vertex ordering is a directed path with probability $2^{-(n-1)}$. Thus μ_n is the expected path count, and Szele's observation [10] gives $P(n) \ge \mu_n$. The expected cycle count is ν_n for $n \ge 3$. The extremal question is how much an orientation can improve on these random benchmarks.

1.1 A quantitative path bound

Divoux and Wang [12] proved that the maximum cycle and path counts are $(L + o(1))\nu_n$ and $(L + o(1))\mu_n$, respectively, where $L = \cosh 1/\cos 1$. Our first result gives a uniform rate of convergence for paths.

Theorem 1.1. There are absolute constants K', n_1 such that every integer $n \ge n_1$ satisfies

$$(L - K'/n)\mu_n \le P(n) \le (L + K'/n)\mu_n.$$

For odd n, the carousel Car_n has vertex set $\mathbb{Z}/n\mathbb{Z}$, with $i \to j$ precisely when the representative of $j - i$ modulo n belongs to $\{1, \dots, (n-1)/2\}$. The lower bound is attained, up to an error of $O(\mu_n/n)$, by odd carousels and by the even tournaments obtained by deleting one vertex from Car_{n+1}.

The issue in the upper bound is uniformity: the error must hold for every tournament, including those with a small set of highly unbalanced vertices. We retain the degree-imbalance penalty while scaling the relevant submatrices and summing the positive path convolution. This yields the same constant for both parities of n.

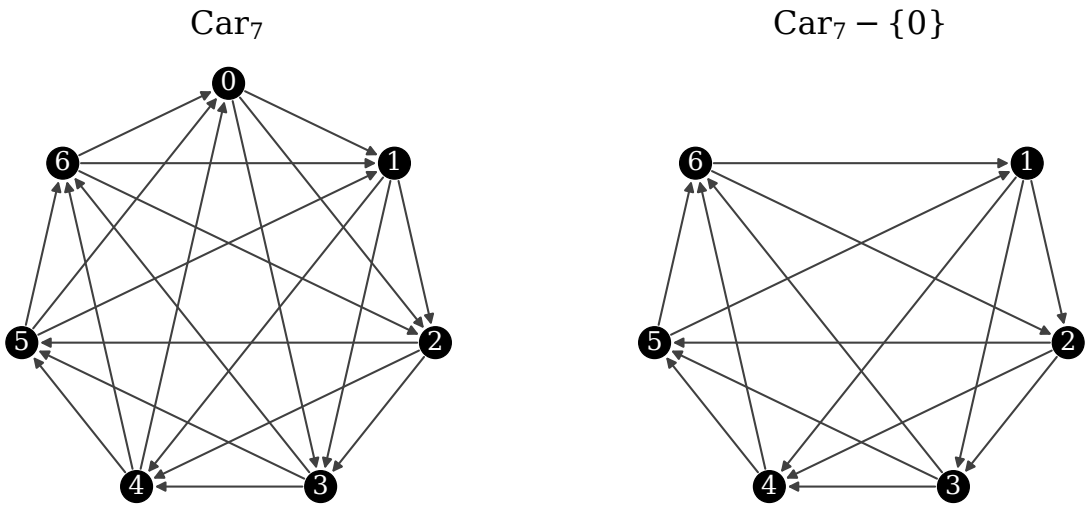


Figure 1: The carousel Car_7 and its even-order subtournament obtained by deleting vertex 0. Every vertex on the left has outdegree 3; those on the right have outdegree 2 or 3. All arcs are shown.

1.2 Degree stability and the quantitative mechanism

The score-sensitive upper estimate behind Theorem 1.1, together with its quantitative lower bound, gives a structural consequence. Write

$$V_{\deg}(T) = \sum_v \left(d^+(v) - \frac{n-1}{2} \right)^2 .$$

Theorem 1.2 (Degree stability). There are absolute constants C, N such that, whenever $n \geq N$, $0 \leq \varepsilon \leq (L-1)/2$ and $H(T)/\mu_n \geq L - \varepsilon$,

$$V_{\deg}(T) \leq C(\varepsilon n^2 + n).$$

In particular, an exact maximizer has $V_{\deg}(T) = O(n)$, or bounded average squared degree deviation. The proof is Corollary 8.3.

This linear scale is necessary at even orders. Every degree deviation is then a half-integer, so $V_{\deg}(T) \geq n/4$, with equality precisely when all outdegrees are $n/2 - 1$ or $n/2$. Hence exact maximizers of even order have $V_{\deg}(T) = \Theta(n)$. This establishes the order in n, not optimal dependence on ε.

The same control holds in a window below the actual maximum. With $\delta_n(T) = (P(n) - H(T))/\mu_n$, Corollary 8.4 gives $V_{\deg}(T) \leq C(\delta_n(T) n^2 + n)$ for $0 \leq \delta_n(T) \leq (L-1)/4$ and sufficiently large n. In particular, a path-count deficit $O(\mu_n/n)$ still gives $V_{\deg}(T) = O(n)$.

Even for exact maximizers, the bound does not give eventual regularity, carousel uniqueness or edit-distance stability: it only bounds the largest degree deviation by $O(\sqrt{n})$, rather than the $o(\sqrt{n})$ required in Theorem 8.1. Divoux–Wang [12, Proposition 8.1] start from a cycle count at least its random benchmark; our hypothesis is a path count near the sharp maximum. These are different structural assumptions.

The technical point is Proposition 7.3. After large variance and exceptional vertices have been excluded, it gives

$$\frac{H(T)}{\mu_n} \leq \rho_n(S) \exp(-\tau/2 + K/n) + B_n, \qquad \tau = \frac{4V_{\deg}(T)}{(n-1)^2}, \quad B_n = o(n^{-1}).$$

Scaling and determinant perturbations produce $K\sqrt{\tau/n}$, which is not $O(n^{-1})$ by itself. The retained negative exponent absorbs it through

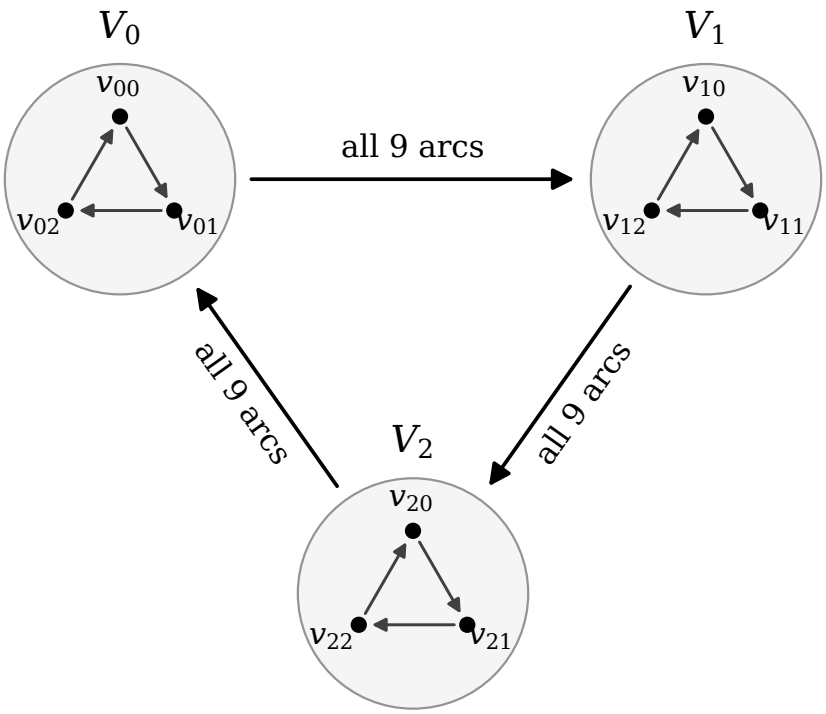


Figure 2: A blow-up of the directed triangle. Each internal block shown is a directed triangle; a thick arrow between blocks represents all 9 arcs in that direction. In the random models, the internal blocks are chosen independently from the specified distributions.

$$K\sqrt{\tau/n} \le \tau/4 + K^2/n.$$

The deletion-size errors are summed against weights with finite fixed moments, instead of being bounded at the largest cutoff. Together these two steps prevent a square-root or logarithmic loss. A quantitative permanent approximation alone does not settle these global errors. Theorems 1.1 and 1.2 are the main counting and structural results; the remaining introductory results describe connections and consequences of the shared spectral ratio.

1.3 Random blow-ups and the angular Hessian

The carousel construction also arises in Wormald's random blow-up models [6]. These models lead to a second question: how does the choice of a fixed base tournament affect the expected Hamiltonian counts?

Fix an odd integer $d \ge 3$ and a tournament T^* on d vertices. Each vertex is replaced by a block of m vertices, and the directions between blocks are prescribed by T^*. In $\mathcal{T}(T^*, m)$ the internal tournaments are independent and uniform; in $\mathcal{R}(T^*, m)$ they are independent and uniform among regular tournaments, so m must be odd. Put $N = dm$. The expected cycle and path counts are normalized by ν_N and μ_N, respectively.

For a regular base, Theorem 4 of Wormald [6] expresses the limiting constant in terms of an angular Hessian, subject to its determinant being nonzero. He computes this determinant for several bases and conjectures that the condition always holds. We prove positive definiteness and evaluate the determinant for every regular base.

Theorem 1.3. Let T^* be regular, with adjacency matrix A. Set $S = A - A^T$, $C = S/d$, and $E = (e_1 - e_d, \dots, e_{d-1} - e_d)$. The Hessian $\mathbf{H}$ of Wormald's angular function at the origin is positive definite, and

$$\mathbf{H} = E^T(I + C^2)(I - C^2)^{-1}E, \qquad \mathcal{H} = d\,\frac{\det(I + C^2)}{\det(I - C^2)} > 0,$$

$$\sqrt{d/\mathcal{H}} = \rho_d(S) := \frac{\det(I + S/d)}{\det(I + iS/d)} \in \mathbb{Q}_{>0}.$$

Here $\mathcal{H} = \det \mathbf{H}$, and the square root is positive. The angular function is defined in Section 9. Together with Theorems 1, 4 and 5 of Wormald [6], this shows that, for fixed d, the normalized expected cycle and path counts in $\mathcal{T}(T^*, m)$ both tend to $\rho_d(S)$, while the corresponding limits in $\mathcal{R}(T^*, m)$ are $e^{1/d}\rho_d(S)$.

Thus the spectral ratio governing the global path bound also determines the blow-up constants. These constants can also be obtained directly from Divoux and Wang [12, Proposition 5.1], without the Hessian calculation; Section 10.1 records that derivation. Their Proposition 7.1 and Lemma 9.1 already give the spectral maximum and the carousel value. The next corollary combines these results with the equality conditions of Fang and Huang [11], the zero-row-sum condition for a regular base, and a separate bound for nonregular bases.

Corollary 1.4. For each fixed odd $d \geq 3$, there exists M_d such that, for every $m \geq M_d$, the carousel Car_d is the unique base, up to isomorphism, maximizing the expected number of Hamiltonian cycles and paths in both models. For $\mathcal{R}(T^*, m)$, only odd m are considered.

The closed form below is the binomial evaluation in Divoux and Wang [12, Section 7 and Lemma 9.1], with the factor $e^{1/d}$ supplied by Wormald [6, Theorem 1]. The known frequencies in Fang and Huang [11, Lemma 4.2] give the following further consequence.

Corollary 1.5. Set

$$r_d = \frac{(1 + 1/d)^d + (1 - 1/d)^d}{(1 + i/d)^d + (1 - i/d)^d}, \qquad \mathfrak{r}_d = e^{1/d} r_d.$$

Along odd integers $d \geq 3$, the sequence $\mathfrak{r}_d$ increases strictly to L.

Theorem 1.3 proves Wormald's Hessian conjecture [6, Conjecture 6]. The limit in Conjecture 2 follows from the known closed form; Corollary 1.5 adds strict monotonicity. The expansion through d^{-5}, with remainder $O(d^{-6})$, is given in the detailed proof companion, §10.4.

1.4 Earlier bounds and related work

The first bounds determined the exponential scale. Szele [10] obtained $P(n) = O(n!2^{-3n/4})$ and conjectured that $(P(n)/n!)^{1/n}$ tends to $1/2$. Alon [2] proved $P(n) = O(n^{3/2}\mu_n)$ by applying Brégman's inequality to the permanent of the adjacency matrix and then transferring the cycle bound to paths by adding a vertex. Friedgut and Kahn [7] improved the polynomial exponent by an entropy argument, giving a cycle bound $O(n^{\alpha}\nu_n)$ with $\alpha \approx 1.249$ and the corresponding path bound.

Lower bounds improve when edge orientations are correlated. Adler, Alon and Ross [9] used an almost complete packing of edge-disjoint triangles, oriented independently as directed triangles, to obtain $(e - o(1))\mu_n$. For a random vertex ordering, the number of packed triangles occurring consecutively tends to Poisson(1), which accounts for the factor e. Wormald's carousel blow-ups [6] gave a cycle lower constant exceeding 2.85588 along an infinite sequence of orders and suggested the limiting value about 2.855958.

The first author's path preprint [14], publicly archived on 8 October 2026, established $(L - O(n^{-1}))\mu_n \leq P(n) \leq (C_* + O(n^{-1}))\mu_n$, with $L < C_*$. It contains the positive path convolution, uniform permanent estimate, nonprincipal scaling and exceptional-vertex analysis used here. The archive precedes the arXiv v1 submission of Divoux and Wang [12] on 16 October. Theorem 1.1 combines that counting framework with the spectral moment comparison of Fang and Huang [11], refining the sharp leading asymptotic in [12, Theorem 1.4] to a uniform normalized error $O(n^{-1})$.

McCullagh [5, §3] already gives an $O(n^{-1})$ determinant approximation; his §4.1 also considers complex perturbations. Divoux–Wang [12, Proposition 3.1 and Remark 2.1] explicitly give a uniform $1+o(1)$ estimate

on fixed bounded-input classes. We treat Theorem 3.1 as an analytic tool with an independent proof and explicit control of parameter dependence. Section 3 compares the hypotheses and determinant factors. We claim neither the first permanent $O(n^{-1})$ approximation nor novelty merely from signed inputs or class-uniform wording. The new quantitative assertion concerns the final path count after scaling, exceptional-vertex removal and summation.

Bhattacharjee's public release of 14 October 2026 [16] contains the full original angular Hessian, its positive definiteness, and the associated determinant results. Our independent Wormald derivation is recorded in the Liu–Ye Git manuscript revision dated 19 October [15] and acknowledged in [13]. These are, respectively, a public-release date and a Git-revision date. The separate single-author path archive [14] concerns the L/C_* counting framework.

1.5 Proof outline and further questions

After fixing notation in Section 2, we develop the permanent and nonprincipal scaling estimates in Sections 3 and 4. The positive path convolution in Section 5 has summable deletion-size moments, allowing these estimates to be combined without a logarithmic loss. Section 6 supplies the all-order spectral comparison. Section 7 separates large variance, exceptional vertices and the remaining class; in the last class, the score penalty in Proposition 7.3 absorbs the square-root error terms. Section 8 gives the matching lower bound and degree-variance stability. Sections 9–11 turn to the angular Hessian and the model extrema: regular noncarousel bases have a strict limiting gap, nonregular bases have normalized expectations tending to zero, and the finite number of base classes at fixed d gives a common threshold. The submission consists of this main text and the Technical appendix, which contains the necessary permanent, scaling and spectral-equality proofs with their original section and equation numbers. The detailed proof companion is separate expanded reading; its elementary calculations and higher-order expansions are optional, not additional proof prerequisites for the main results.

The optimal error order, the first correction coefficient and the exact path maximizers at each finite order remain open. Another question is to estimate the growth of M_d. Letting the base and block orders grow together would require bounds uniform in d, beyond the fixed-base limits used here.

2. Preliminaries and notation

2.1 The random baseline and error conventions

The random path benchmark is $\mu_n = n!/2^{n-1}$, as derived in Section 1. Let $\mathcal{T}_n$ denote all tournaments with vertex labels $[n]$. In this paper, a uniform $O(n^{-1})$ error means

$$\exists K < \infty\, \exists n_0\, \forall n \geq n_0\, \forall T \in \mathcal{T}_n : \quad |\text{error}(n,T)| \leq K/n.$$

The constants K, n_0 are uniform over tournaments and deletion sets. They may depend on additional fixed parameters b, A_0, B_0, when present.

2.2 Sign matrices and spectral factors

Definition 2.1 (Sign matrices and scores).

The adjacency matrix A has zero diagonal and $A_{ij} = 1$ precisely when $i \to j$. Define

$$S = A - A^{\mathsf{T}}, \qquad 2A = J - I + S, \qquad s = S\mathbf{1}.$$

Here I is the identity matrix in the current dimension, $J = \mathbf{1}\mathbf{1}^{\mathsf{T}}$, and $\mathbf{1}$ is the all-ones column vector. In $I_p, J_p, \mathbf{1}_p$, the subscript explicitly specifies the dimension p. The superscript T denotes transpose. If d_i^+ is the outdegree, then $s_i = 2d_i^+ - (n-1)$ and $\sum_i s_i = 0$. The Euclidean and maximum vector norms are

$$\|x\|_2 = \left(\sum_i |x_i|^2\right)^{1/2}, \quad \|x\|_\infty = \max_i |x_i|.$$

We use the notation

$$\mathcal{E} = \|s\|_2^2, \qquad \Delta_s = \|s\|_\infty, \qquad a = \frac{s}{n-1}, \qquad \tau = \|a\|_2^2.$$

The normalized scores a and τ are defined for $n \geq 2$. An odd-order tournament is regular when every vertex has outdegree $(n-1)/2$, equivalently $s = 0$. An even-order tournament is balanced when every outdegree is $n/2$ or $n/2 - 1$, equivalently $s_i \in \{-1, 1\}$. Thus Δ_s measures the largest vertex imbalance, whereas τ measures the total squared imbalance after normalization.

Definition 2.2 (Submatrices and permanents).

For a vertex set U, the notation $A[U] = A[U, U]$ denotes the principal submatrix. For possibly different row and column sets R, C, the submatrix $A[R, C]$ selects those rows and columns, always in increasing original vertex-label order. The same convention applies to a general matrix $M[R, C]$. The complement U^c is taken in the current full vertex set. Empty sums equal zero and empty products equal one; both the determinant and the permanent of the empty matrix equal one. For a square matrix of order m, denoted by M, the permanent is

$$\operatorname{per} M = \sum_{\pi \in \mathfrak{S}_m} \prod_{i=1}^{m} M_{i,\pi(i)}.$$

Definition 2.3 (Frequencies and spectral factors).

The singular values of a real matrix M are the nonnegative square roots of the eigenvalues of $M^{\mathsf{T}}M$. The operator norm $\|M\|_{\mathrm{op}} = \sup_{\|x\|_2=1} \|Mx\|_2$ equals the largest singular value. The square of the Frobenius norm $\|M\|_F = (\sum_{i,j} |M_{ij}|^2)^{1/2}$ is the sum of the squared singular values. For the skew-symmetric matrix S, write $\pm i\lambda_j$ for its nonzero eigenvalue pairs, where $i^2 = -1$ and $\lambda_j > 0$, adding zero frequencies when needed. Each λ_j occurs twice among the singular values. Define

$$x_j = \frac{\lambda_j^2}{n^2}, \qquad D_n(S) = \det(I - S^{\mathsf{T}}S/n^2)^{-1/2},$$

$$\rho_n(S) = D_n(S) \det(I + S/n) = \prod_j \frac{1 + x_j}{1 - x_j}.$$

The skew-symmetric singular values occur in pairs, and $\|S\|_F^2 = n(n-1)$, so

$$\sum_j x_j = \frac{n-1}{2n}, \qquad \|S/n\|_{\mathrm{op}}^2 \leq \frac{n-1}{2n} < \frac{1}{2}.$$

Therefore D_n and ρ_n are positive and bounded by absolute constants. In particular,

$$D_n(S) = \det(I + iS/n)^{-1}.$$

These identities follow by decomposing S into real skew-symmetric 2×2 blocks and using $\|S\|_F^2 = n(n-1)$.

For $n = 1$, we have $S = [0]$, the frequency sum is empty, and $D_1(S) = \rho_1(S) = 1$.

Every log below is a natural logarithm. Using $-\log(1-x) = \int_0^x (1-u)^{-1}\,du \le 2x$ for $0 \le x \le 1/2$, we obtain

$$0 \le \log D_n(S) \le 2\sum_j x_j < 1, \qquad 1 \le D_n(S) < e.$$

Similarly, $\log(1+x) \le x$ gives $\log \rho_n(S) < 3/2$. Multiplication by $D_n(S)$ or $\rho_n(S)$ therefore preserves the order of the error. Lemma 6.1 gives a sharper bound.

3. A uniform permanent approximation

Balancing a submatrix changes both its entries and its dimension. The permanent approximation must therefore have constants controlled by fixed density and spectral-gap bounds, rather than by the particular matrix. The theorem below provides this uniformity for the deletions used in Section 4. Its proof uses the coefficient and tail estimates established in **Technical appendix, §§3.2–3.6**, with the original numbering retained; those estimates are part of the proof.

Role and comparison with earlier work. Theorem 3.1 is a self-contained version of the permanent tool needed below. We do not use its wording as a separate novelty claim. In the nonnegative case set $P = J_n/n$, $B = E/n$ and $D = P + B$. The hypotheses imply

$$\frac{1}{n^2}\sum_{i,j} |E_{ij}|^p \le C^p \quad (p \ge 1), \qquad D^{\mathsf T}D = P + B^{\mathsf T}B.$$

Thus fixed entry bounds give weak boundedness, and the singular-value bound gives a fixed gap (also for $D^{\mathsf T}D$). McCullagh's [5, §3] factor becomes exactly

$$\det(I + P - D^{\mathsf T}D)^{-1/2} = \det(I - B^{\mathsf T}B)^{-1/2} = \det(I - BB^{\mathsf T})^{-1/2}.$$

Class-uniformity alone also does not distinguish a theorem valid for every admissible sequence: if no common $O(n^{-1})$ constants existed for fixed C, q, one could choose increasing n_j with $n_j|\mathrm{error}(E_{n_j})| \to \infty$ and fill other dimensions with $E_n = 0$, contradicting that sequence assertion. This argument concerns the stated fixed class, not a claim of constants uniform over all weakly bounded classes.

Our independent proof tracks constants through contracted non-Gaussian cores, a coefficient window growing with n, and a complex coefficient tail. These estimates make the tool reusable after deletion and scaling. We do not assert that an identified gap in McCullagh's proof is repaired here. Signed inputs alone are not a priority distinction, since [5, §4.1] already discusses complex perturbations. The graph-counting applications use only nonnegative inputs.

Theorem 3.1 (Uniform permanent approximation).

Fix $0 \le C < \infty$ and $0 \le q < 1$. Let $n \ge 1$, and suppose $E \in \mathbb{R}^{n\times n}$ satisfies

$$E\mathbf{1} = E^{\mathsf T}\mathbf{1} = 0, \qquad \max_{i,j} |E_{ij}| \le C, \qquad \|E/n\|_{\mathrm{op}} \le q.$$

Put $B = E/n$. Uniformly over all matrices satisfying these conditions,

$$\frac{\mathrm{per}(J_n + E)}{n!} = \det(I - BB^{\mathsf T})^{-1/2} + O_{C,q}(n^{-1}). \tag{3.1}$$

Equation (3.1) can also be written as the determinant factor times the relative error $1 + O_{C,q}(n^{-1})$. The subscript C, q means that the error constant and the threshold dimension may depend on these two parameters fixed in advance, but must apply simultaneously to every admissible matrix.

Proof. Choose fixed $1 < \sigma < R < 1/q$ (omit the upper restriction if $q = 0$), and put

$$f(z) = \frac{\operatorname{per}(J_n + zE)}{n!}, \qquad G(z) = \det(I - z^2 BB^{\mathsf{T}})^{-1/2}, \qquad \mathcal{G}_{C,q}(r) = \exp\left(\frac{C^2 r^2}{2(1 - q^2 r^2)}\right).$$

Since $\|B\|_F^2 \le C^2$, $1 \le G(1) \le \mathcal{G}_{C,q}(1)$ and $|G(z)| \le \mathcal{G}_{C,q}(|z|)$ on the fixed disk. The positive square-root branch is determined at zero.

Separate inclusion–exclusion for row and column indices expresses coefficients as bipartite multigraph sums. Double centering eliminates degree-one vertices. Degree-two components are exactly the Gaussian pairings and sum to G. For the remaining components, contract degree-two chains **before taking absolute values**: the alternating products of B, B^{T} retain the spectral gap. A chain of length $\ell \ge 2$ has contraction bounded by $C^2 q^{\ell-2}/n$. The residual core is classified by excess (edges minus vertices), giving one factor n^{-1} per excess. The exact labeling compensation and the core activity bound are proved in Technical appendix §§3.2–3.3. This order of contraction preserves the operator-norm control.

For some fixed $\alpha > 0$, set $M = \lfloor \alpha n \rfloor$. The finite coefficient window, including restoration of $n^k/(n)_k$, has error $K_1/n + K_2/n^2$ (Technical appendix §3.4). Its deviation is of order k^2/n, and the Gaussian coefficients are summed with a fixed radius greater than one to absorb this weight. Permanent polarization and the positive-semidefinite permanent estimate give $\max_{|z|=R} |f(z)| \le \exp(\beta\sqrt{n})$, up to a fixed factor, where $\beta = RC/(1 - Rq)$; Cauchy's estimate then controls the coefficients above M (Technical appendix §3.5). Together these bounds yield

$$|f(1) - G(1)| \le \frac{K_1}{n} + \frac{K_2}{n^2} + K_3 e^{\beta\sqrt{n} - \alpha n \log R} + K_4 e^{-\alpha n \log \sigma}.$$

Every constant and the threshold depend only on C, q and the radii fixed from them. Both exponential terms are $o(n^{-1})$. Finitely many smaller orders are covered using $|f(1)| \le (1 + C)^n$ and $G(1) \le \mathcal{G}_{C,q}(1)$. This proves the uniform absolute bound; $G(1) \ge 1$ gives the relative form. □

Technical appendix §3.1 gives a signed example with fixed parameters at arbitrarily large orders; §3.6 records the two estimates used to quantify the nonnegative proof of [12]. The applications use the stated nonnegative specialization.

4. Scaling and nonprincipal submatrices

Deleting different row and column sets disturbs both the marginals and the total mass. We must balance the remaining matrix while tracking the scalar factor needed to recover its permanent.

Write $a = S\mathbf{1}/(n-1)$, $\tau = \sum a_i^2$, $P_p = J_p/p$ and $\Pi_p = I_p - P_p$. For a matrix X of order p, let $\mathfrak{m}(X) = \sum X_{ij}$ and $g(X) = (X\mathbf{1} - \mathbf{1}, X^{\mathsf{T}}\mathbf{1} - \mathbf{1})$. The normalization $C = 2A/(n-1)$ has marginals $\mathbf{1} + a$ and $\mathbf{1} - a$. The three estimates below control balancing, its logarithmic cost, and the Gaussian factor. Their parameter dependence is the input to the uniform bounds in Lemmas 4.4–4.5.

4.1 Scaling and Gaussian factors

Lemma 4.1 (Local scaling).

Let $p \geq 1$, and let X have total mass p and be a real $p \times p$ matrix. Write

$$\alpha = X\mathbf{1}_p - \mathbf{1}_p, \qquad \beta = X^{\mathsf{T}}\mathbf{1}_p - \mathbf{1}_p, \qquad \varepsilon = \max(\|\alpha\|_\infty, \|\beta\|_\infty).$$

Assume $|X_{ij}| \leq C/p$, that the sum of absolute values in each row and each column is at most 2, and that

$$X_0 = X - \frac{\alpha\mathbf{1}_p^{\mathsf{T}} + \mathbf{1}_p\beta^{\mathsf{T}}}{p}, \qquad E = X_0 - P_p$$

satisfies $|E_{ij}| \leq C_0/p$ and $\|E\|_{\mathrm{op}} \leq q < 1$, where $C, C_0 \geq 0$ and $0 \leq q < 1$ are fixed parameters. Define

$$L_0 = \frac{3}{2} + C_0 + \frac{C_0^2}{1-q},$$

$$\varepsilon_0 = \min\left\{\frac{1}{256L_0^2}, \frac{1-q}{2(32L_0+2)}\right\}.$$

If $\varepsilon \leq \varepsilon_0$, there exist finite real vectors x, y with $\sum_i x_i = \sum_j y_j$ such that every row sum and column sum of $B = \mathrm{diag}(e^x)X\,\mathrm{diag}(e^y)$ equals 1. Moreover,

$$\|(x,y)\|_\infty \leq 4L_0\varepsilon, \qquad |B_{ij}| \leq 2C/p, \qquad \|B - P_p\|_{\mathrm{op}} \leq (1+q)/2.$$

If X is entrywise nonnegative, then B is doubly stochastic and has the same zero pattern as X.

Lemma 4.2 (Displacement and capacity).

Under the preceding hypotheses, assume in addition that $X \geq 0$. Let $\theta_X = \sum_i x_i + \sum_j y_j$ denote the total scaling potential. There is a constant depending only on the fixed density and spectral-gap parameters such that

$$\|(x,y)\|_2 \leq K\|(\alpha,\beta)\|_2, \qquad \|B - X\|_F \leq \frac{K}{\sqrt{p}}\|(\alpha,\beta)\|_2,$$

$$0 \leq \theta_X \leq K\|(\alpha,\beta)\|_2^2.$$

General mass. For a nonnegative matrix X of arbitrary positive total mass, suppose that $\widetilde{X} = pX/\mathfrak{m}(X)$ satisfies the hypotheses of Lemma 4.1, and apply the preceding conclusions to $\widetilde{X}$. If B is its scaling, the following exact identities hold:

$$\theta_X = \theta_{\widetilde{X}} + p\log\frac{p}{\mathfrak{m}(X)}, \qquad \mathrm{per}\, X = e^{-\theta_X}\,\mathrm{per}\, B.$$

In particular, if $\kappa = \mathfrak{m}(X) - p$, then $\mathrm{per}\, X \leq e^{\kappa}\,\mathrm{per}\, B$. When the mass differs from p, θ_X itself need not be nonnegative.

Proof of the scaling estimates. Fix the gauge $\sum x_i = \sum y_j$. The reference matrix H_0 has diagonal blocks I and off-diagonal blocks X_0, X_0^{T}, while the actual linearization H_X has diagonal blocks $\mathrm{diag}(X\mathbf{1}), \mathrm{diag}(X^{\mathsf{T}}\mathbf{1})$ and off-diagonal blocks X, X^{T}. Put $u = (\mathbf{1}, -\mathbf{1})$ and $N = uu^{\mathsf{T}}/(2p)$ to fix the gauge. On the zero-sum subspaces, the inverse of $\mathsf{H}_0 + N$ is expressed through $(I - EE^{\mathsf{T}})^{-1}$ and $(I - E^{\mathsf{T}}E)^{-1}$. The dense entry bound and operator gap give its induced 1- and ∞-norms at most L_0. Both corresponding norms of $\mathsf{H}_X - \mathsf{H}_0$ are at most 3ε; hence the Neumann series bounds the inverse of $\mathsf{H}_X + N$ by $2L_0$. The stated threshold then makes the nonlinear map a contraction in the radius $4L_0\varepsilon$ ball. This proves existence

and the stated entry and spectral bounds. The block inverse and contraction estimates are supplied in Technical appendix §4.1.

For nonnegative X, apply this construction to the convex potential

$$\Phi_X(x,y) = \sum_{ij} X_{ij} e^{x_i+y_j} - \sum x_i - \sum y_j.$$

Along the segment to the balanced point, its Hessian on the gauge subspace has a positive lower bound depending only on the fixed parameters. Integration gives the two-norm bound, and the entry bound gives the Frobenius displacement. Comparing the potential at zero and at its minimum gives $0 \le \theta_X \le K\|g(X)\|_2^2$. Scalar normalization proves the general-mass identity and $-\theta_X \le \kappa$. In particular, the nonnegativity of the potential cost is used at total mass p; the general-mass argument uses the one-sided bound $-\theta_X \le \kappa$.

Define $\mathcal{G}(Z) = \det(I - Z^{\mathsf{T}}Z)^{-1/2}$ for $\|Z\|_{\mathrm{op}} < 1$; zero padding and changing to ZZ^{T} preserve its value.

Lemma 4.3 (Gram deletion and stability).

Suppose $\|Z\|_{\mathrm{op}} \le q_* < 1$. Delete the row set I and the column set J, leaving the sets R, T. Then

$$0 \le \log \mathcal{G}(Z) - \log \mathcal{G}(Z[R,T]) \le \frac{\sum_{i\in I}\|Z_{i,\cdot}\|_2^2 + \sum_{j\in J}\|Z_{\cdot,j}\|_2^2}{2(1-q_*^2)}.$$

Double centering. At order $m \ge 1$, assume $\|W\|_{\mathrm{op}} \le q_*$ for a square retained matrix W. Set $u = \mathbf{1}_m/\sqrt{m}$. Then

$$0 \le \log \mathcal{G}(W) - \log \mathcal{G}(\Pi_m W \Pi_m) \le \frac{\|W^{\mathsf{T}}u\|_2^2 + \|Wu\|_2^2}{2(1-q_*^2)}.$$

Stability. For two real matrices of the same dimensions whose operator norms are both at most q_*, we also have

$$|\log \mathcal{G}(U) - \log \mathcal{G}(V)| \le \frac{\|U\|_F + \|V\|_F}{2(1-q_*^2)}\|U - V\|_F.$$

Proof. Integrate the derivative of $-\frac{1}{2}\log\det(I-H)$ along the positive-semidefinite Gram loss caused by deleting rows. Its trace is the sum of the deleted squared row norms. Switch Gram sides for column deletion and apply the same argument to the constant-direction projections. The stability inequality follows along the Gram segment from $\|U^{\mathsf{T}}U - V^{\mathsf{T}}V\|_* \le (\|U\|_F + \|V\|_F)\|U - V\|_F$. □

4.2 Uniform deletion bounds

Lemma 4.4 (Nonprincipal permanent bound).

Fix $0 \le b < 1$ and positive constants A_0, B_0. For a tournament core of order $N \ge 2$, set $a = S\mathbf{1}/(N-1)$ and $\tau = \sum_i a_i^2$. Let $I, J \subseteq [N]$ be the deleted row and column sets, respectively. Here I, J are sets, distinguished from the identity and all-ones matrices by the set operations below. Assume

$$\max_i |a_i| \le b, \qquad \tau \le A_0 \log N, \qquad |I| = |J| = t \le B_0 \log N.$$

Define $\ell_i = (1+a_i)^{-1}$, $r_i = (1-a_i)^{-1}$, and

$$\Gamma = \prod_i (1-a_i^2), \qquad \mathcal{M}_\tau = \frac{\tau+\tau^2}{N} + \frac{\tau^{3/2}}{\sqrt{N}}.$$

Let $R = [N] \setminus I$ and $T = [N] \setminus J$ be the retained row and column sets, respectively; here T is a column index set. Then

$$\operatorname{per} A[R,T] \le e^{-1} D_N(S)\Gamma \left(\prod_{i\in I}\ell_i\right)\left(\prod_{j\in J} r_j\right)\frac{(N-t)!}{2^{N-t}}\exp(\varepsilon_{\tau,t}),$$

$$\varepsilon_{\tau,t} \le K\left[\mathcal{M}_\tau + \sqrt{\tau/N} + \frac{t+1}{N} + t\sqrt{\tau/N} + \frac{t^2}{N}\right].$$

For every sufficiently large N, this estimate holds simultaneously for every admissible tournament and every pair I, J. The constant K and the threshold depend only on b, A_0, B_0.

Proof. Use $n = N$ and $m = n - t$. Perform paired score correction and mass normalization:

$$\begin{aligned} C &= \frac{2A}{n-1}, \qquad & C' &= \operatorname{diag}(\ell) C \operatorname{diag}(r),\\ \nu &= \mathfrak{m}(C') - n, & \widehat{C} &= \frac{nC'}{n+\nu},\\ X &= \frac{n}{m}\widehat{C}[R,T], & \kappa &= \mathfrak{m}(X) - m. \end{aligned}$$

Put $v_i = a_i/(1-a_i^2)$, $w_i = a_i^2/(1-a_i^2)$, $V = \sum v_i$, $W = \sum w_i$. The cancellation of the first-order mass change is recorded by the exact identity

$$\nu = \frac{W^2 - V^2 + W + 2w^{\mathsf{T}} S v}{n-1}, \qquad |\nu| \le K\mathcal{M}_\tau.$$

If e, f are the marginal errors of $\widehat{C}$, deleting the independent row and column sets gives

$$\kappa = \frac{n}{m}\left[-\sum_{i\in I} e_i - \sum_{j\in J} f_j + \sum_{i\in I, j\in J}\widehat{C}_{ij} - \frac{t^2}{n}\right]. \tag{4.1}$$

Normalize $\widetilde{X} = mX/(m+\kappa)$. The preprocessing estimates in Technical appendix §4.3 give, uniformly,

$$|\kappa| \le K(t\sqrt{\tau/n} + t^2/n), \quad \|g(\widetilde{X})\|_\infty \le K(\sqrt{\tau/n} + t/n), \quad \|g(\widetilde{X})\|_2 \le K(\sqrt{\tau} + t/\sqrt{n}).$$

The entries of $\widetilde{X}$ lie in $[0, K/m]$ and its doubly centered kernel has operator norm at most $1/\sqrt{2} + o(1)$. Fixing the gap $q = 3/4$ therefore makes Lemma 4.1 applicable above a common threshold. Its scaling B_X has entries $O(1/m)$, with centered norm eventually at most $4/5$. Lemmas 4.2–4.3 and the same preprocessing give

$$-\theta_X \le \kappa, \qquad \log \mathcal{G}(B_X - P_m) \le \log D_n(S) + K\big(\sqrt{\tau/n} + (t+1)/n\big).$$

The exact restoration is

$$\operatorname{per} A[R,T] = \Gamma \prod_{i\in I}\ell_i \prod_{j\in J} r_j \left[\frac{m(n-1)(n+\nu)}{2n^2}\right]^m e^{-\theta_X} \operatorname{per} B_X. \tag{4.2}$$

Theorem 3.1 gives $\operatorname{per} B_X = (m!/m^m)\mathcal{G}(B_X - P_m)(1+O(m^{-1}))$. The remaining scalar is $(m!/2^m)(1-1/n)^m(1+\nu/n)^m$, whose logarithmic deviation from $e^{-1}m!/2^m$ is $O((t+1)/n + |\nu|)$. Combining these bounds proves the displayed error exponent. All gaps, densities and thresholds were fixed from b, A_0, B_0, which establishes the stated uniform quantifiers. □

Lemma 4.5 (Permanent approximation for small scores).

Suppose $\Delta_s = \|S\mathbf{1}_n\|_\infty = o(\sqrt{n})$. Fix $0 < B_0 < \infty$, delete any $t \le B_0 \log n$ rows and t columns, and write $m = n - t$. Uniformly over these choices,

$$\operatorname{per} A[R,T] = e^{-1} D_n(S) \frac{m!}{2^m} \left[1 + O_{B_0}\left(\frac{(\Delta_s + t + 1)^2}{n} \right) \right]. \tag{4.3}$$

More precisely, for fixed $B_0 > 0$ there are constants $C_{\mathrm{err}}, \delta > 0$ and an integer $N_0 \geq 4$ such that the following holds uniformly for $n \geq N_0$. Let S be a tournament sign matrix and let $I, J \subseteq [n]$ satisfy $|I| = |J| = t \leq B_0 \log n$. Put $\Delta_s = \|S\mathbf{1}_n\|_\infty$ and $\eta = (\Delta_s + t + 1)^2/n$. If $\eta \leq \delta$, the absolute relative error in equation (4.3) is at most $C_{\mathrm{err}}\eta$. Here $R = [n] \setminus I$, $T = [n] \setminus J$, and $m = n - t$; the two deleted sets may be chosen independently. In the asymptotic range of the lemma, $\eta \to 0$.

Proof. Take $X = (n/m)C[R,T]$ directly, without paired score correction. Equation (4.1) and the marginal identities give

$$|\kappa| = O(t(\Delta_s + t)/n), \quad \|g(X)\|_\infty = O((\Delta_s + t)/n), \quad \|g(X)\|_2 = O((\Delta_s + t)/\sqrt{n}).$$

After normalizing to mass m, Lemmas 4.1–4.2 give $|\theta_X| = O((\Delta_s + t)^2/n)$ and scaling displacement $O((\Delta_s + t)/n)$. To compare the full Gaussian factors, set $Y = (S - I)/(n - 1)$ and use skew-symmetry to obtain

$$Y^{\mathsf{T}}Y = \frac{S^{\mathsf{T}}S + I}{(n-1)^2}, \qquad Y^{\mathsf{T}}Y - \frac{S^{\mathsf{T}}S}{n^2} \succeq 0.$$

The last positive-semidefinite difference has trace $O(n^{-1})$, since $\operatorname{tr}(S^{\mathsf{T}}S) = n(n-1)$. Integrating the log-determinant derivative across this Gram difference, with a fixed spectral gap, gives $|\log \mathcal{G}(Y) - \log D_n(S)| = O(n^{-1})$. Combining it with Lemma 4.3, deletion, centering and scalar normalization gives

$$|\log \mathcal{G}(B_X - P_m) - \log D_n(S)| = O(\eta).$$

The restoration identity is now $\operatorname{per} A[R,T] = [m(n-1)/(2n)]^m e^{-\theta_X} \operatorname{per} B_X$. Insert Theorem 3.1 and $(1 - 1/n)^m = e^{-1} \exp(O((t+1)/n))$. The normalized ratio is $e^u(1+v)$ with $|u| \leq K\eta$, $|v| \leq K/n \leq K\eta$. Choose a fixed sufficiently small δ and use $|e^u - 1| \leq e^{|u|}|u|$ to obtain the claimed finite, uniform relative-error bound. □

5. Path convolution and a determinant representation

5.1 Positive path convolution

Lemma 5.1 (Path convolution).

For every tournament,

$$H(T) = \sum_{U \subseteq V(T)} \det(I + A[U]) \operatorname{per} A[U^c]. \tag{5.1}$$

Proof.

The permanent of the adjacency matrix counts directed cycle covers. The convolution equation (5.1) converts cycle-cover counts on complementary vertex sets into the path count. It is the tournament specialization of Proposition 2 of Irving and Omar [1], where the complementary adjacency matrix is $\overline{A} = J - A = I + A^{\mathsf{T}}$ and includes the diagonal entries.

For a direct generating-function proof, set $X = \operatorname{diag}(z_1, \ldots, z_n)$ and give each visit to vertex i a factor z_i. The walk generating function is

$$1 + \mathbf{1}^{\mathsf{T}}(I - XA)^{-1}X\mathbf{1} = \frac{\det(I + X\overline{A})}{\det(I - XA)}.$$

The equality follows from the rank-one determinant lemma. Extracting the coefficient of $z_1 \cdots z_n$ selects exactly the Hamiltonian paths. The trace of a matrix is the sum of its diagonal entries,

$$\operatorname{tr} M = \sum_i M_{ii}.$$

For the denominator, the formal identity $\det(I - XA)^{-1} = \exp(\sum_{r\geq 1} \operatorname{tr}((XA)^r)/r)$ shows that the coefficients of squarefree monomials, in which each variable has exponent zero or one, count disjoint directed cycle covers and thus equal permanents. Combining this with the principal-minor expansion of the numerator gives

$$H(T) = \sum_U \det \overline{A}[U] \operatorname{per} A[U^c].$$

Substituting $\overline{A}[U] = I + A[U]^{\mathsf{T}}$ proves the identity.

Write $T[U]$ for the subtournament induced by the vertex set U, and let $k = |U|$. The symmetric part of $I + A[U]$ is the positive definite matrix $(I + J)/2$. Thus its real eigenvalues are positive, its nonreal eigenvalues occur in conjugate pairs, and its determinant is positive. The squared row norms are $1 + d_i^+(T[U])$, whose mean is $(k + 1)/2$. Hadamard's inequality and the arithmetic–geometric mean inequality give

$$0 < \det(I + A[U]) \leq h_k, \qquad h_k = \left(\frac{k+1}{2}\right)^{k/2}.$$

This proves the convolution and positivity claims. □

Set $h_0 = 1$ and define

$$w_k = \frac{2^k h_k}{k!}.$$

Lemma 5.2 (Fixed moments and a uniform tail). For every fixed $c > 0$ and integer $r \geq 0$,

$$\sum_{k\geq 0} (k+1)^r c^k w_k < \infty.$$

For sufficiently large n, put $k_n = \lceil 4 \log n / \log\log n \rceil$. There is an absolute constant C such that the normalized contribution of $|U| > k_n$ to (5.1) is bounded uniformly by

$$B_n := \frac{C}{2}\sqrt{n+1} \sum_{k>k_n} w_k = n^{-3/2+o(1)} = o(n^{-1}).$$

Proof. Stirling's formula gives $\log(c^k w_k) = -(k/2)\log k + O_c(k)$, and the exact ratio of successive terms is at most $K_c/\sqrt{k+1}$. Hence every fixed moment converges and $\sum_{k>k_n} c^k w_k = n^{-2+o(1)}$. Brégman's bound and degree balancing [2, Lemma 2.1 and Corollary 2.3] give per $A_T \leq C\sqrt{m+1}\, m!/2^m$ for every tournament of order m; enlarge C to include $m = 0$. Consequently, the normalized convolution terms of size k are at most

$$\frac{\binom{n}{k} h_k C\sqrt{n-k+1}\,(n-k)!\, 2^{-(n-k)}}{n!\, 2^{-(n-1)}} \leq \frac{C}{2}\sqrt{n+1}\, w_k.$$

Summing proves the assertion. □

We call $|U| \leq k_n$ the short terms. In the error estimates below, retaining the fixed moments of $c^k w_k$ instead of bounding k by k_n avoids logarithmic losses.

5.2 The emergence of the spectral factor

First suppose $\Delta_s = o(\sqrt{n})$. By Lemma 4.5, for $|U| = k \le k_n$ we have

$$\operatorname{per} A[U^c] = e^{-1} D_n(S) \frac{(n-k)!}{2^{n-k}} \left(1 + O\left(\frac{(\Delta_s + k + 1)^2}{n}\right)\right).$$

For integers $0 \le k \le n$, define the falling factorial by

$$(n)_k = \prod_{h=0}^{k-1} (n-h),$$

Thus $(n)_0 = 1$, and $(n)_k > 0$ in this range. The normalization satisfies $((n-k)!/2^{n-k})/\mu_n = 2^{k-1}/(n)_k$. Substituting the leading term into Lemma 5.1 therefore gives the normalized sum

$$\frac{e^{-1} D_n(S)}{2} \sum_{|U| \le k_n} \frac{2^{|U|}}{(n)_{|U|}} \det(I + A[U]),$$

The relative error is uniform over tournaments and sets $|U| \le k_n$. Replacing $(n)_k$ by n^k in the short sum and adding the tail yields the principal-minor identity below; Section 8.1 estimates the resulting errors.

$$\sum_U (2/n)^{|U|} \det(I + A[U]) = \det\left(I + \frac{2}{n}(I + A)\right).$$

Since $2A = J - I + S$, the rank-one term J asymptotically contributes a factor of two, while the scalar term contributes e. More precisely, the small-score assumption gives

$$\det\left(I + \frac{2}{n}(I + A)\right) = 2e \det(I + S/n)\left(1 + O(n^{-1} + \Delta_s^2/n^2)\right).$$

The factor e^{-1} in the permanent estimate comes from restoring $(1 - 1/n)^{n-k}$ and reflects the zero diagonal. The factor $1/2$ comes from the path normalization above. They cancel the factor $2e$ in the generating determinant, leaving $D_n(S) \det(I + S/n) = \rho_n(S)$. The two determinants have separate origins: the first approximates cycle-cover counts, and the second sums the weights in the path convolution. Theorem 8.1 will complete the error summation and prove

$$\frac{H(T)}{\mu_n} = \rho_n(S) + O((\Delta_s + 1)^2/n) \qquad \text{when } \Delta_s = o(\sqrt{n}).$$

To obtain an upper bound for all tournaments, Section 7 separately handles large score variance and extreme degrees, retaining the score penalty in the remaining class. Section 6 bounds this spectral factor, culminating in the all-order comparison of Proposition 6.6.

6. The spectral bound and equality

The spectral-ratio bound in this section is due to Divoux and Wang [12, Proposition 7.1], using the comparison of Fang and Huang [11]. The determinant polynomial and the carousel switching are also in [12, Section 7 and Lemma 9.1]. We give the details and follow the equality conditions needed for the fixed-base corollary.

6.1 The spectral radius and an elementary uniform bound

Lemma 6.1 (Spectral cap). Every tournament sign matrix of order n satisfies

$$\|S\|_{\mathrm{op}} \le \cot\left(\frac{\pi}{2n}\right) < \frac{2n}{\pi}, \qquad \rho_n(S) \le \frac{3n-1}{n+1} < 3.$$

Proof. The phase-ordering argument [4, Theorem 3.1] gives the operator bound as follows. Apply a simultaneous signed permutation to z and S so that the phases of the nonzero entries of z lie in increasing order in $[0,\pi)$. Then $c_{ij} = \mathrm{Im}(\overline{z}_i z_j) \ge 0$ for $i<j$, and, for the transitive sign matrix T_n^0,

$$z^* i S z = -2\sum_{i<j} S_{ij} c_{ij} \le 2\sum_{i<j} c_{ij} = z^*(-iT_n^0)z.$$

Rayleigh's principle reduces the bound to T_n^0. Subtracting consecutive row eigenvalue equations gives $(\lambda-1)v_i = (\lambda+1)v_{i+1}$. The geometric ratio $r = (\lambda-1)/(\lambda+1)$ satisfies $r^n = -1$ by the first row equation. Thus the eigenvalues are $i\cot((2j-1)\pi/(2n))$, $1 \le j \le n$. Their largest modulus is $\cot(\pi/(2n))$, and $\tan u > u$ gives the strict inequality (the case $n=1$ is immediate).

For the ratio bound, write the eigenvalues in nonzero pairs $\pm i\lambda_j$ and put $x_j = \lambda_j^2/n^2$. The Frobenius identity gives $\sum_j x_j = (n-1)/(2n) < 1/2$. With $\phi(x) = \log((1+x)/(1-x))$, the derivative $\phi'(x) = 2/(1-x^2)$ is increasing on $[0,1)$ and $\phi(0)=0$. Hence $\phi(a+b) \ge \phi(a)+\phi(b)$ for $a,b \ge 0$, $a+b<1$. Repeatedly merging the x_j gives

$$\log \rho_n(S) = \sum_j \phi(x_j) \le \phi\left(\frac{n-1}{2n}\right),$$

which proves the stated bound, including $n=1$.

The eigenvalue calculation also yields

$$\det(I + zT_n^0) = \frac{(1+z)^n + (1-z)^n}{2}.$$

Indeed, each even transitive principal determinant equals one: the positive cotangent frequencies pair with their reciprocals at complementary angles. Odd principal determinants vanish by skew-symmetry. The principal-minor expansion is therefore $\sum_h \binom{n}{2h} z^{2h}$. □

Historical remark. Combining the stronger cap $x_j \le 4/\pi^2$ with $\sum_j x_j \le 1/2$ gives the earlier constant

$$C_* = \frac{3\pi^4 + 4\pi^2 - 32}{\pi^4 + 4\pi^2 - 32} > L.$$

It is the relaxed convex maximum attained at $4/\pi^2, 1/2 - 4/\pi^2, 0, \ldots$. The present proof uses the elementary bound $\rho_n < 3$ to exclude exceptional vertices and the all-order sharp comparison in Proposition 6.6 for the final upper bound.

6.2 Product comparison and equality

Theorem 6.2 (Fang–Huang).

We use the following form of Fang–Huang, Corollary 3.4. Define the comparison matrix K_d by

$$(K_d)_{ij} = \begin{cases} 1, & i+j \le d, \\ 0, & i+j = d+1, \\ (-1)^{i+j-d-2}, & i+j \ge d+2. \end{cases}$$

Let $p \ge 1$, let $x \in \mathbb{R}^d$, and let $S_1, \dots, S_p$ be tournament sign matrices of order d. Write x^* for the vector obtained by arranging the absolute values of the coordinates of x in decreasing order. Then

$$(S_1 \cdots S_p x)^* \prec_w K_d^p x^*.$$

The right-hand side is a nonnegative decreasing vector. Weak majorization $u \prec_w v$ means

$$\sum_{i=1}^{r} u_i \le \sum_{i=1}^{r} v_i \quad (1 \le r \le d).$$

Moreover, vector equality in the outermost comparison implies equality for every suffix product. To apply this to Frobenius norms, we first record the squared-norm inequality under weak majorization, including its equality case.

Lemma 6.3 (Truncated sums under weak majorization).

Write $(a-t)_+ = \max(a-t, 0)$. For nonnegative decreasing vectors satisfying $u \prec_w v$ and $t \ge 0$,

$$\sum_{i=1}^{d} (u_i - t)_+ = \max_{0 \le r \le d} \left(\sum_{i=1}^{r} u_i - rt \right) \le \max_{0 \le r \le d} \left(\sum_{i=1}^{r} v_i - rt \right) = \sum_{i=1}^{d} (v_i - t)_+.$$

Proof.

The first equality follows by taking $r = \#\{i : u_i > t\}$: adding coordinates at most t, or removing coordinates greater than t, cannot increase the sum. For $a \ge 0$, we also have

$$2 \int_0^\infty (a-t)_+ \, dt = 2 \int_0^a (a-t) \, dt = 2[at - t^2/2]_0^a = a^2.$$

Integrating coordinate by coordinate gives

$$\|u\|_2^2 = 2 \int_0^\infty \sum_i (u_i - t)_+ \, dt \le 2 \int_0^\infty \sum_i (v_i - t)_+ \, dt = \|v\|_2^2.$$

Suppose the squared norms are equal. The function $g(t) = \sum_i (v_i - t)_+ - \sum_i (u_i - t)_+$ is continuous and nonnegative, with integral zero. If $g(t_0) > 0$, then $g(t) > g(t_0)/2$ on an interval of positive length, a contradiction. Thus $g \equiv 0$. Comparing right derivatives gives

$$\#\{i : u_i > t\} = \#\{i : v_i > t\} =: N(t).$$

For coordinate j,

$$u_j = \int_0^\infty \mathbf{1}_{\{\#\{i : u_i > t\} \ge j\}} \, dt = \int_0^\infty \mathbf{1}_{\{N(t) \ge j\}} \, dt = v_j.$$

Equality of squared norms therefore forces $u = v$. Taking $x = e_j$ and using $e_j^* = e_1$ yields

$$\|S^p e_j\|_2^2 \le \|K_d^p e_1\|_2^2.$$

Equality also forces $(S^p e_j)^* = K_d^p e_1$.

6.3 Moments of the comparison matrix

The following calculation works at either parity. Let T_d have all upper-triangular entries equal to 1, let R_0 reverse the coordinates, and put $D_0 = \mathrm{diag}(1, -1, \dots)$. Let Λ list the odd indices increasingly, followed by the even indices decreasingly, and set $U = D_0\Lambda$. The entry formula for K_d gives

$$T_d = \Lambda^T D_0 K_d R_0 \Lambda, \qquad T_d^2 = -U^T K_d^2 U, \qquad U e_1 = e_1.$$

Indeed, after this ordering every upper-triangular entry of the first matrix is 1; the second identity follows from $K_d^T = K_d$ and $T_d^T = -T_d$. The signed cyclic shift in Technical appendix A.2 commutes with T_d, so all diagonal entries of T_d^{2p} coincide. Hence

$$\|T_d^p\|_F^2 = (-1)^p \operatorname{tr} T_d^{2p} = d(-1)^p (T_d^{2p})_{11} = d\|K_d^p e_1\|_2^2.$$

Summing the columnwise consequence of Lemma 6.3 gives, for every $p \ge 1$,

$$\|S^p\|_F^2 \le d\|K_d^p e_1\|_2^2 = \|T_d^p\|_F^2.$$

6.4 The extremum of the spectral ratio

Lemma 6.4 (The spectral ratio and equality).

Let S be a tournament sign matrix of order d, and put $r_d = \rho_d(T_d)$. Then

$$\rho_d(S) \le r_d, \quad \text{with equality if and only if } S = \Omega^T T_d \Omega.$$

Here Ω is a signed permutation matrix.

Proof.

Let S be any tournament sign matrix of order d, with nonzero eigenvalues $\pm i\lambda_j$, where $\lambda_j > 0$, and put $x_j = \lambda_j^2/d^2$. Since $2\sum_j \lambda_j^2 = \|S\|_F^2 = d(d-1)$, we have $x_j \le (d-1)/(2d) < 1/2$. In the skew-symmetric canonical form, a zero block contributes 1 to both determinants, while a nonzero two-dimensional block contributes $1 + x_j$ and $1 - x_j$, respectively. Hence

$$\det(I + S/d) = \prod_j (1 + x_j), \qquad \det(I + iS/d) = \prod_j (1 - x_j) > 0.$$

For $0 \le x < 1$, integrate the geometric series:

$$\begin{aligned} \log\frac{1+x}{1-x} &= \int_0^x \left(\frac{1}{1+t} + \frac{1}{1-t}\right) dt \\ &= 2\int_0^x \frac{dt}{1-t^2} = 2\sum_{r=0}^{\infty} \int_0^x t^{2r}\, dt = 2\sum_{r=0}^{\infty} \frac{x^{2r+1}}{2r+1}. \end{aligned}$$

The interchange follows from uniform convergence of the geometric series on $[0, x]$. The skew-symmetric canonical form also gives $\|S^p\|_F^2 = 2\sum_j \lambda_j^{2p}$. Substituting term by term yields

$$\log \rho_d(S) = 2 \sum_j \sum_{r=0}^{\infty} \frac{\lambda_j^{4r+2}}{(2r+1) d^{4r+2}}$$
$$= \sum_{r=0}^{\infty} \frac{\|S^{2r+1}\|_F^2}{(2r+1) d^{4r+2}} \le \sum_{r=0}^{\infty} \frac{\|T_d^{2r+1}\|_F^2}{(2r+1) d^{4r+2}} = \log \rho_d(T_d).$$

Put $r_d = \rho_d(T_d)$. If equality holds above, every nonnegative difference in the series vanishes. The term $r = 1$ gives $\|S^3\|_F^2 = \|T_d^3\|_F^2$. Since this difference is itself the sum of the nonnegative column differences, equality holds in each column. The equality criterion of Section 6.2 and the suffix equality in Fang–Huang's theorem then give

$$(S^3 e_j)^* = K_d^3 e_1, \qquad (S^2 e_j)^* = K_d^2 e_1.$$

Take squared norms in the second identity and sum over j:

$$\operatorname{tr} S^4 = \|S^2\|_F^2 = d \|K_d^2 e_1\|_2^2 = \|T_d^2\|_F^2 = \operatorname{tr} T_d^4.$$

The fourth-moment classification in Technical appendix A.3 shows that equality is equivalent to $S = \Omega^T T_d \Omega$ for a signed permutation matrix Ω. Conversely, $\Omega^T \Omega = I$ implies

$$I + tS = \Omega^T (I + tT_d) \Omega, \qquad \det(I + tS) = (\det \Omega)^2 \det(I + tT_d) = \det(I + tT_d).$$

Take $t = 1/d$ and $t = i/d$ to obtain $\rho_d(S) = r_d$, proving the converse. $\square$

If S is also regular, equality holds if and only if its tournament is isomorphic to Car_d. Indeed, write $\Omega = D_\varepsilon P$, where P is a permutation matrix and $D_\varepsilon = \operatorname{diag}(\varepsilon_1, \dots, \varepsilon_d)$ with $\varepsilon_i \in \{-1, 1\}$. Put $\varepsilon = (\varepsilon_1, \dots, \varepsilon_d)^T$. Then

$$S = P^T D_\varepsilon T_d D_\varepsilon P, \qquad P S P^T = D_\varepsilon T_d D_\varepsilon.$$

Relabel by P and continue to write S for the resulting matrix. If the tournament is regular, then $S\mathbf{1} = 0$, and $D_\varepsilon^2 = I$ gives

$$D_\varepsilon T_d \varepsilon = 0, \qquad T_d \varepsilon = 0.$$

The difference between rows i and $i + 1$ has entry 1 in each of columns $i, i + 1$ and zero elsewhere. Therefore

$$0 = (T_d \varepsilon)_i - (T_d \varepsilon)_{i+1} = \varepsilon_i + \varepsilon_{i+1}.$$

It follows that $\varepsilon_i = (-1)^{i-1} \varepsilon_1$. The common sign ε_1 cancels in $D_\varepsilon T_d D_\varepsilon$, leaving the alternating switching. Technical appendix A.2 identifies this tournament with Car_d. Thus, among regular tournaments, equality holds only for the carousel, up to isomorphism.

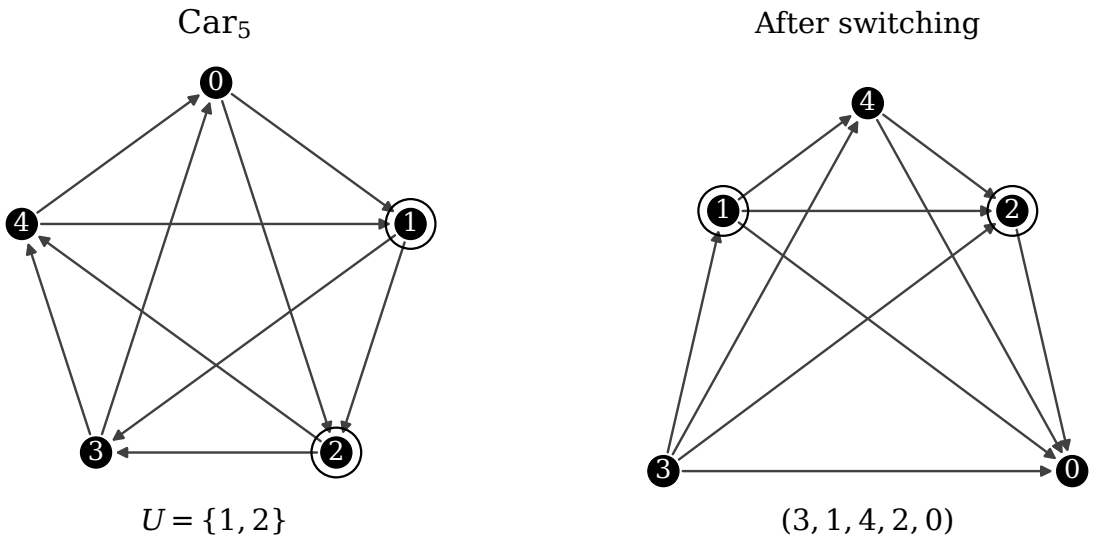


Figure 3: Reversing all arcs between $U = \{1, 2\}$ and its complement in Car_5 gives the transitive order $(3, 1, 4, 2, 0)$. Circled vertices belong to U. This switching conjugates the sign matrix as $S \mapsto DSD$ and preserves the spectral ratio; it need not preserve the Hamiltonian path count.

6.5 A gap from the sixth moment

Lemma 6.5 (Strict sixth-moment deficit).

If regular S is not ordinarily isomorphic to the carousel, the preceding equality argument gives

$$\Delta_3 := \|T_d^3\|_F^2 - \|S^3\|_F^2 > 0.$$

Proof.

Modulo 2, $S \equiv J + I$. Since $J^2 = dJ$ and $J^3 = d^2J$,

$$S^3 \equiv (J + I)^3 = I + 3J + 3dJ + d^2J \equiv I + (1 + d + d^2)J \equiv I + J \pmod 2.$$

The last step uses that $d(d + 1)$ is even. Also $(S^3)^T = -S^3$, so its diagonal entries vanish and its off-diagonal entries are odd. If $z = 2k + 1$,

$$z^2 = 4k(k + 1) + 1 \equiv 1 \pmod 8.$$

Pair the skew-symmetric positions in the squared norm:

$$\|S^3\|_F^2 = 2\sum_{i<j}((S^3)_{ij})^2 \equiv 2\binom{d}{2} = d(d - 1) \pmod{16}.$$

This also applies to T_d. Thus Δ_3 is a positive multiple of 16, and $\Delta_3 \geq 16$. Retaining just term $r = 1$ in the logarithmic series gives

$$\log \frac{r_d}{\rho_d(S)} \geq \frac{\Delta_3}{3d^6} \geq \frac{16}{3d^6}.$$

Exponentiate and rearrange:

$$\rho_d(S) \leq r_d e^{-16/(3d^6)}, \qquad r_d - \rho_d(S) \geq r_d(1 - e^{-16/(3d^6)}) > 0.$$

The gap depends on the base order d. Section 11.2 obtains a corresponding block threshold by comparing the limits.

6.6 The sharp comparison at every order

Proposition 6.6 (All-order spectral comparison). For every tournament sign matrix S of order n, $\rho_n(S) \le r_n$, where $r_n = L + O(n^{-1})$ is the explicit quotient below. The $O(n^{-1})$ constant is absolute and uniform over tournaments.

Proof. Section 6.3 gives the moment comparison at every order, without regularity. Substituting it into the absolutely convergent series of Section 6.4 yields

$$\log \rho_n(S) = \sum_{r\ge 0} \frac{\|S^{2r+1}\|_F^2}{(2r+1)n^{4r+2}} \le \log \rho_n(T_n).$$

The determinant polynomial in Lemma 6.1 also holds at both parities. Hence

$$\rho_n(S) \le r_n = \frac{(1+1/n)^n + (1-1/n)^n}{(1+i/n)^n + (1-i/n)^n} = L + O(n^{-1}).$$

For the last step, the Taylor estimate for the logarithm gives, for $z \in \{1, -1, i, -i\}$, $n\log(1+z/n) = z + O(n^{-1})$, while the denominator tends to $2\cos 1 > 0$. The quotient therefore has an $O(n^{-1})$ error. These estimates depend on the order alone, not on the tournament. □

7. The upper bound for arbitrary tournaments

Write $V_{\text{deg}}(T)$ for degree variance, reserving $V(T)$ for the vertex set:

$$V_{\text{deg}}(T) = \sum_i \left(d_i^+ - \frac{n-1}{2}\right)^2 = \frac{1}{4}\|S\mathbf{1}\|_2^2, \qquad \tau = \frac{4V_{\text{deg}}(T)}{(n-1)^2}.$$

The proof has three cases. If $V_{\text{deg}}(T) \ge 16n^2\log n$, the variance penalty in Brégman's inequality makes the path count negligible. Otherwise, vertices with degrees outside $[(n-1)/20, 19(n-1)/20]$ number at most $O(\log n)$. If any such vertices exist, separating them from the rest gives a path count below the random expectation. In the remaining tournaments, normalized scores stay a fixed distance from ± 1, and $\tau = O(\log n)$. For them, the permanent estimate below retains the factor $e^{-\tau}$ and controls the final error by $O(n^{-1})$.

The constant K in this section may increase from one occurrence to the next; after the degree thresholds above are fixed, it is absolute. Section 5.1 has already bounded the long terms by $o(n^{-1})\mu_n$, so here we estimate only short terms $|U| \le k_n$.

The required precision differs across the three classes: a fixed gap below L suffices for the first two, while the last requires the precise main term. The parameters in the scaling lemmas are fixed before taking a common large-order threshold.

Proposition 7.1 (Uniform sharp upper bound). There are absolute constants K, n_0 such that every tournament T of order $n \ge n_0$ satisfies

$$H(T) \le (L + K/n)\mu_n.$$

7.1 The role of nonprincipal deletions

The independent row and column deletions in Lemma 4.4 are needed for the cross-block matchings in Section 7.3: an edge entering the core occupies a column, whereas an edge leaving it occupies a row. The occupied sets can differ. In Section 7.4 the deletions are principal, and the same estimate supplies a common score product that remains outside the path convolution. These two uses explain why a principal-submatrix estimate alone would be insufficient.

7.2 Large score variance

Lemma 7.2 (Permanent bound with a variance penalty).

There is an absolute constant K such that, for every order $m \geq 1$, a tournament Q with outdegrees d_i and degree variance $V_{\deg}(Q) = \sum_i (d_i - (m-1)/2)^2$ satisfies

$$\operatorname{per} A_Q \leq K\sqrt{m+1}\frac{m!}{2^m}\exp\left(-\frac{V_{\deg}(Q)}{8m^2}\right).$$

Proof.

Concavity of the row-degree factor strengthens the Brégman bound [2]. If some $d_i = 0$, the adjacency matrix has a zero row and its permanent is zero. Otherwise, define $f(k) = \log(k!)/k$ for positive integers. Direct calculation gives, for $k \geq 2$,

$$2f(k) - f(k-1) - f(k+1) = \frac{2[\log k - f(k-1)] - k\log(1+1/k)}{k(k+1)}.$$

Applying the arithmetic–geometric mean inequality to $1, \ldots, k-1$ gives $\log k - f(k-1) \geq \log 2$. Also, $(1+1/k)^k < e < 3$, so the numerator is at least $\log(4/3) \geq 1/4$. Consequently, on the degree interval $[1, m-1]$, the piecewise linear interpolation of the integer-point function $f(k) + k^2/(8m^2)$ is concave. Jensen's inequality at the mean $\eta = (m-1)/2$ gives

$$\sum_i f(d_i) \leq m\widetilde{f}(\eta) - \frac{V_{\deg}(Q)}{8m^2} + \frac{1}{32m},$$

Here $\widetilde{f}$ is the linear interpolation. The last term handles a half-integer mean and can be omitted when the mean is an integer. Stirling's formula gives

$$\exp(m\widetilde{f}(\eta)) \leq K\sqrt{m+1}\frac{m!}{2^m}.$$

Combining this with Brégman's row-degree bound proves the variance estimate.

We apply this bound to the tournament associated with each short deleted set. We must therefore check that its variance remains sufficiently large after deletion.

After deleting a size-k set U, each remaining centered degree changes by at most $k/2$, and the sum of the original squared deviations at deleted vertices is at most $kn^2/4$. Applying Cauchy–Schwarz to the cross term gives

$$V_{\deg}(T-U) \geq V_{\deg}(T) - \frac{kn^2}{4} - k\sqrt{nV_{\deg}(T)}.$$

If $V_{\deg}(T) \geq 16n^2\log n$, then uniformly for every $k \leq k_n$,

$$V_{\deg}(T-U) \geq (1-o(1))V_{\deg}(T).$$

Thus the variance penalty in every short permanent is at most $n^{-2+o(1)}$, absorbing the preceding $\sqrt{n}$ factor. Summation with the summable determinant weights bounds the total normalized short contribution by $n^{-3/2+o(1)}$. Adding the long-deletion tail gives, uniformly over this graph class,

$$V_{\deg}(T) \geq 16n^2 \log n \quad \Longrightarrow \quad H(T)/\mu_n = o(1/n).$$

Concavity and deletion estimates. We record the calculations supporting the two bounds above. Write $L_k = \log(k!)$. Then

$$\begin{aligned} 2f(k) - f(k-1) - f(k+1) &= \frac{2(L_{k-1} + \log k)}{k} - \frac{L_{k-1}}{k-1} - \frac{L_{k-1} + \log k + \log(k+1)}{k+1} \\ &= \frac{2\log k - 2L_{k-1}/(k-1) - k\log(1+1/k)}{k(k+1)}. \end{aligned}$$

AM–GM gives

$$((k-1)!)^{1/(k-1)} \leq \frac{1 + \cdots + (k-1)}{k-1} = k/2,$$

Hence $\log k - f(k-1) \geq \log 2$. Also, $k\log(1+1/k) < 1 < \log 3$, and

$$\log(4/3) = \int_1^{4/3} \frac{du}{u} \geq \frac{1/3}{4/3} = \frac{1}{4}.$$

The second difference is therefore at least $1/[4k(k+1)] \geq 1/(4m^2)$. Set $h(k) = f(k) + k^2/(8m^2)$. Then

$$2h(k) - h(k-1) - h(k+1) \geq \frac{1}{4m^2} - \frac{2}{8m^2} = 0.$$

If η is an integer, $\widetilde{h}(\eta) = \widetilde{f}(\eta) + \eta^2/(8m^2)$. If $\eta = k + 1/2$, then

$$\frac{k^2 + (k+1)^2}{2} = \eta^2 + \frac{1}{4}, \quad \widetilde{h}(\eta) = \widetilde{f}(\eta) + \frac{\eta^2 + 1/4}{8m^2}.$$

Jensen's inequality therefore gives

$$\begin{aligned} \sum_i f(d_i) + \frac{\sum_i d_i^2}{8m^2} = \sum_i h(d_i) &\leq m\widetilde{h}(\eta) \\ &\leq m\widetilde{f}(\eta) + \frac{m\eta^2}{8m^2} + \frac{1}{32m}, \\ \sum_i f(d_i) &\leq m\widetilde{f}(\eta) - \frac{\sum_i (d_i - \eta)^2}{8m^2} + \frac{1}{32m}. \end{aligned}$$

The factorial estimate for both parities is verified at the end of this proof.

Write $b_i = d_i^+ - (n-1)/2$. For a retained vertex, let e_i count its outgoing edges to U. Then

$$b_i' = d_i^+ - e_i - (n-k-1)/2 = b_i + \delta_i, \qquad \delta_i = k/2 - e_i, \quad |\delta_i| \leq k/2.$$

Thus

$$\begin{aligned} V_{\deg}(T-U) &= \sum_{i\notin U}(b_i+\delta_i)^2 \\ &\geq V_{\deg}(T) - \sum_{i\in U} b_i^2 - 2\left|\sum_{i\notin U} b_i\delta_i\right| \\ &\geq V_{\deg}(T) - kn^2/4 - k\sum_{i\notin U}|b_i| \\ &\geq V_{\deg}(T) - kn^2/4 - k\sqrt{nV_{\deg}(T)}. \end{aligned}$$

When $V \geq 16n^2\log n$, the relative losses are respectively at most

$$\frac{k}{64\log n} = o(1), \qquad k\sqrt{n/V} \leq \frac{k}{4\sqrt{n\log n}} = o(1).$$

It follows that $V_{\deg}(T-U)/(8(n-k)^2) \geq (2-o(1))\log n$, so the penalty is $n^{-2+o(1)}$. Multiplying by $\sqrt{n}$ and $\sum_k w_k < \infty$ gives $n^{-3/2+o(1)}$.

The factorial factor. Use Stirling's formula $\log(m!) = (m+1/2)\log m - m + c + O(m^{-1})$. With c denoting its constant,

$$f(r) = \log r - 1 + \frac{\log r}{2r} + \frac{c}{r} + O(r^{-2}).$$

First take $m = 2r+1$, so that the average outdegree is the integer r. Multiplication by m gives

$$mf(r) = m\log r - m + \left(1+\frac{1}{2r}\right)\log r + 2c + \frac{c}{r} + O(r^{-1}).$$

The logarithm of the natural scale is

$$\log(m!/2^m) = m\log(m/2) - m + \frac{1}{2}\log m + c + O(m^{-1}).$$

Denote their difference by Δ_m:

$$\begin{aligned} \Delta_m &= m\log\frac{2r}{m} + \log r - \frac{1}{2}\log m + c + O(\log m/m) \\ &= m\log(1-1/m) + \frac{1}{2}\log m - \log 2 + \log(1-1/m) + c + O(\log m/m). \end{aligned}$$

Since $m\log(1-1/m) = -1 + O(1/m)$ and all other terms except $\frac{1}{2}\log m$ are uniformly bounded, $\Delta_m \leq \frac{1}{2}\log m + K$.

Next take $m = 2r$ with $r \geq 2$. The mean is $r - 1/2$, and linear interpolation gives

$$m\widetilde{f}((m-1)/2) = r[f(r-1) + f(r)].$$

Substitution term by term yields

$$\begin{aligned} r[f(r-1)+f(r)] = {} & r\log(r-1) + r\log r - 2r \\ & + \frac{r}{2(r-1)}\log(r-1) + \frac{1}{2}\log r \\ & + c\frac{r}{r-1} + c + O(r^{-1}). \end{aligned}$$

Subtracting $\log((2r)!/2^{2r}) = 2r\log r - 2r + \frac{1}{2}\log(2r) + c + O(r^{-1})$ gives

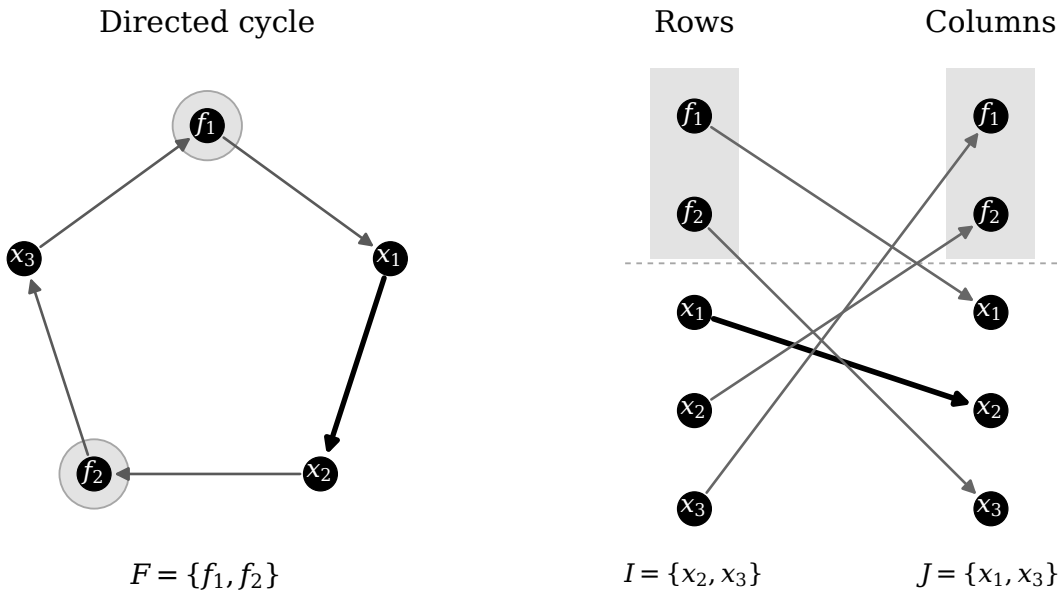


Figure 4: A directed five-cycle in a permanent term and its row-column matching. Shaded vertices form the exceptional set $F = \{f_1, f_2\}$. Cross-block arcs occupy core rows $I = \{x_2, x_3\}$ and core columns $J = \{x_1, x_3\}$, leaving the nonprincipal entry $A[x_1, x_2]$ shown in bold. Only the arcs of this permanent term are drawn.

$$\Delta_{2r} = r\log(1-1/r) + \frac{r}{2(r-1)}\log(r-1) - \frac{1}{2}\log 2 + c\frac{r}{r-1} + O(r^{-1}).$$

The first term is nonpositive. The second satisfies

$$\frac{r}{2(r-1)}\log(r-1) \le \frac{1}{2}\log r + \frac{\log r}{2(r-1)} \le \frac{1}{2}\log r + \frac{1}{2},$$

Here $\log r \le r-1$. The constant term $cr/(r-1)$ has absolute value at most $2|c|$. Thus again $\Delta_m \le \frac{1}{2}\log m + K$. Exponentiating in both parity cases gives

$$\exp(m\widetilde{f}((m-1)/2)) \le e^K\sqrt{m}\,m!/2^m.$$

The finitely many values of m outside the expansion range are covered by increasing the same K. □

7.3 Exceptional vertices

Assume from now on that $V_{\deg}(T) < 16n^2\log n$. Define the exceptional set

$$F = \left\{ i : \frac{d_i^+}{n-1} \notin [1/20, 19/20] \right\}, \qquad f = |F|, \qquad M = T - F, \qquad N = n - f.$$

Each exceptional vertex contributes to $V_{\deg}(T)$ at least a fixed positive multiple of n^2, so $f = O(\log n)$. A nonexceptional vertex originally has absolute score at most $0.9(n-1)$, and deleting F changes its score by at most f. Consequently, uniformly for all sufficiently large n,

$$\max_i |(S_M\mathbf{1})_i/(N-1)| < 0.95, \qquad \tau_M = O(\log n).$$

For the squared score sum, use $\|S_M\mathbf{1}\|_2 \le \|S\mathbf{1}\|_2 + f\sqrt{N}$ and divide by $N-1$. The same argument remains valid after further deleting a short core subset. Throughout these deletions the original exceptional set F is retained, allowing the constants to be chosen uniformly.

Cross-block matchings. Classify cycle covers according to their connections between F and the core M, thereby estimating per A_T. An exceptional vertex has few neighbors in one direction. A cycle cover must use edges in both directions unless that vertex is matched within F, and the latter choice incurs a cost of order $1/N$. Use the paired core weights ℓ, r from Lemma 4.4, whose total deviation satisfies

$$\sum_{i\in M} |\ell_i - 1| + \sum_{i\in M} |r_i - 1| \le K\sqrt{N\log n}.$$

For $x \in F$, write $q_x = N^{-1}|\{j \in M : x \to j\}|$. Its distance from $[0, 1/20] \cup [19/20, 1]$ is $O(f/n)$. The weighted sums of cross-edge neighbors on the two sides are bounded by $N(q_x + \beta)$ and $N(1 - q_x + \beta)$, where $\beta = O(\sqrt{\log n/n})$. Set

$$u_x = 2(q_x + \beta), \qquad v_x = 2(1 - q_x + \beta).$$

Uniformly for every x, we may choose

$$u_x v_x \le c_n = 19/100 + o(1), \qquad u_x, v_x \le L_n = 2 + o(1).$$

Decompose the permanent according to the permutation edges lying inside F. If there are s internal edges, their row set R and column set C both have size s, and there are exactly $t = f - s$ cross edges in each direction. The deleted core row and column sets I, J both have size t, but need not coincide. The exact four-block expansion is

$$\operatorname{per} A_T = \sum_{s=0}^{f} \sum_{\substack{R,C\subseteq F\\ |R|=|C|=s}} \sum_{\substack{I,J\subseteq M\\ |I|=|J|=f-s}} \operatorname{per} A[R, C]\ \operatorname{per} A[F\setminus R, J]\ \operatorname{per} A[I, F\setminus C]\ \operatorname{per} A[M\setminus I, M\setminus J].$$

Every contributing permutation belongs to exactly one class. First apply the uniform bound of Lemma 4.4 to the core permanent and then separate the two cross-edge sums. Deleting a core row contributes ℓ_i, and deleting a core column contributes r_j. For fixed R, C, the outgoing cross-edge sum is exactly

$$\begin{aligned}&\sum_{\substack{J\subseteq M\\ |J|=t}} \operatorname{per} A[F\setminus R, J] \prod_{j\in J} r_j\\ &= \sum_{\phi: F\setminus R\hookrightarrow M} \prod_{x\in F\setminus R} A_{x,\phi(x)} r_{\phi(x)} \le \prod_{x\in F\setminus R} \sum_{j\in M} A_{xj} r_j.\end{aligned}$$

Independently, the incoming cross-edge sum is

$$\begin{aligned}&\sum_{\substack{I\subseteq M\\ |I|=t}} \operatorname{per} A[I, F\setminus C] \prod_{i\in I} \ell_i\\ &= \sum_{\psi: F\setminus C\hookrightarrow M} \prod_{y\in F\setminus C} A_{\psi(y),y} \ell_{\psi(y)} \le \prod_{y\in F\setminus C} \sum_{i\in M} A_{iy}\ell_i.\end{aligned}$$

Here $\hookrightarrow$ means an injection. Each injection is counted exactly once by its image set and its corresponding permanent term, with no extra factor $t!$. The image sets I, J need neither coincide nor be disjoint: they belong to the row and column copies of the core, respectively. Since every factor is nonnegative, the injectivity restrictions may be dropped. The product of the two upper bounds is

$$N^{2t}2^{-2t}\left(\prod_{x\notin R}u_x\right)\left(\prod_{y\notin C}v_y\right).$$

The permanent inside the exceptional set is at most $s!$. After division by $n!/2^n$, the full factor restoring factorials and powers of two is

$$2^s N^{2t}\frac{(N-t)!}{(N+f)!}=\left(\frac{2}{N}\right)^s\exp(O(f^2/N)).$$

Here $N+f-(N-t)=f+t=2f-s$, so the factorial ratio contributes $N^{-(2f-s)}\exp(O(f^2/N))$. Multiplication by N^{2t} leaves exactly N^{-s}.

Let $h=|R\cap C|$. The paired product satisfies

$$\left(\prod_{x\notin R}u_x\right)\left(\prod_{y\notin C}v_y\right)\le c_n^{f-2s}L_n^{2s}(c_n/L_n^2)^h\le c_n^{f-2s}L_n^{2s}.$$

The first inequality follows by counting vertices carrying two factors, one factor or no factor; it remains valid even when $f-2s$ is negative. For $t\le f=O(\log n)$, the error in Lemma 4.4 is uniformly $o(1)$, and its factor $\Gamma\le 1$ may be discarded. Using $\binom{f}{s}^2 s!\le f^{2s}/s!$, the full sum gives

$$\frac{\operatorname{per}A_T}{n!/2^n}\le(1+o(1))e^{-1}D_N(S_M)c_n^f\exp(O(f^2/N))\exp\left(\frac{2L_n^2f^2}{c_n^2N}\right).$$

All additional exponents here are $o(1)$. More explicitly, after deleting a short core subset U, the core order is $N'=n-f-|U|\sim n$, meaning $N'/n\to 1$, while the exceptional set remains the original F. The normalized core scores are still eventually at most 0.95, their squared sum is $O(\log n)$, and the cross-edge deletion size is $t\le f=O(\log n)$. Hence the error exponent in Lemma 4.4 is

$$O\left(\frac{(\log n)^{3/2}}{\sqrt{n}}+\frac{(\log n)^2}{n}\right)=o(1),$$

The constants are independent of U,I,J. The cross-edge neighbor proportions change from those in the original graph by only $O((f+|U|)/n)$, and the total one-norm deviation of the weights remains $O(\sqrt{n\log n})$. Thus the cross-edge sum estimates above also hold uniformly. The core Gaussian factor becomes $\exp(O(|U|/n))$ times the original one: applying the Gaussian row-and-column deletion lemma to S_M/N gives logarithmic loss $O(|U|/n)$, and changing the normalization scale from N to $N-|U|$ has the same order of cost.

In the path convolution, the proportion of size-k subsets intersecting F is at most fk/n. The general permanent bound therefore bounds their full normalized short contribution by

$$K\frac{f}{\sqrt{n}}\sum_k kw_k=o(1).$$

For subsets contained in the core, restore the falling factorial and sum the positive principal minors. For every core order $N\le n$, the rank-one determinant identity gives

$$\det\left(I_N+\frac{2}{n}(I_N+A_M)\right)\le 2e\det(I_N+S_M/N).$$

To prove this, first factor out $(1+1/n)^N\le e$. The remaining all-ones rank-one factor is at most $1+N/(n+1)<2$, because the real quadratic form of the inverse of $I+S_M/(n+1)$ is at most the squared

vector norm. Finally, the skew-frequency determinant product increases monotonically as the scale increases from $1/(n+1)$ to $1/N$.

Restoring the falling factorial for short subsets contributes $\exp(O(k_n^2/n)) = 1 + o(1)$. Since eventually $c_n < 1/4$, the uniform estimates above give

$$\frac{H(T)}{\mu_n} \le (1/4)^f \rho_N(S_M) + o(1) \le 3/4 + o(1) < 1, \qquad f \ge 1.$$

Since $P(n) \ge \mu_n$, every maximizing tournament of sufficiently large order has no exceptional vertex. A weaker bound for short convolution terms avoiding F is $(10/3)4^{-f}(1+\varepsilon_n)$. Denote the combined contribution of the other short terms and the long terms by δ_n. Then

$$\frac{H(T)}{\mu_n} \le \frac{10}{3} 4^{-f}(1+\varepsilon_n) + \delta_n, \qquad \varepsilon_n, \delta_n \longrightarrow 0. \tag{7.1}$$

Both errors tend to zero uniformly over the low-variance class. Take n large enough that $1+\varepsilon_n \le 21/20$ and that the short terms meeting F and the long terms each contribute at most $1/25$. For $f \ge 1$, equation (7.1) gives

$$\frac{10}{3} \cdot \frac{1}{4} \cdot \frac{21}{20} + \frac{1}{25} + \frac{1}{25} = \frac{191}{200} < 1.$$

Both estimates suffice for the upper bound. The calculations below verify the uniform bounds used in this reduction, including the core weights, the matching multiplicities, and the common large-order threshold.

Core and weight bounds. An exceptional vertex satisfies

$$|d_i^+ - (n-1)/2| > \frac{9}{20}(n-1), \quad f\frac{81}{400}(n-1)^2 \le V_{\deg}(T) < 16n^2 \log n,$$

Hence $f < (6400/81)(n/(n-1))^2 \log n$. For the core,

$$\max_i |s_i^M| \le 0.9(n-1) + f, \quad \frac{0.9(n-1)+f}{n-f-1} \longrightarrow 0.9,$$

$$\tau_M \le \frac{2\|s\|_2^2 + 2f^2 N}{(N-1)^2} = O(\log n) + O((\log n)^2/n) = O(\log n).$$

Since $|\ell_i - 1| \le |a_i|/(1-b)$ and $|r_i - 1| \le |a_i|/(1-b)$,

$$\sum_i (|\ell_i - 1| + |r_i - 1|) \le \frac{2}{1-b} \sum_i |a_i| \le \frac{2}{1-b}\sqrt{N\tau_M} = O(\sqrt{N \log n}).$$

This bounds the difference between the weighted and unweighted count of any neighbor subset by the total deviation above. Specifically,

$$\sum_j A_{xj} r_j \le N q_x + K\sqrt{N \log n}, \quad \sum_i A_{ix} \ell_i \le N(1-q_x) + K\sqrt{N \log n}.$$

For $q \in [0, 1/20] \cup [19/20, 1]$, we have $4q(1-q) \le 4(1/20)(19/20) = 19/100$. The $O(f/n)$ perturbation of the neighbor proportion and the perturbation $\beta = o(1)$ together give $u_x v_x \le 19/100 + o(1)$.

In the four-block expansion, $t = f - s$. There are s edges within the exceptional set, t from the exceptional set to the core, t from the core to the exceptional set, and $N - t$ within the core. Their total is

$$s + t + t + (N - t) = N + f = n.$$

There are no signs in a permanent expansion, and each permutation occurs once after classification. The full powers-of-two and factorial factor is

$$\frac{[(N-t)!/2^{N-t}]\,[N^{2t}/2^{2t}]}{(N+f)!/2^{N+f}} = 2^{f-t}N^{2t}\frac{(N-t)!}{(N+f)!} = 2^{s}N^{2t}\frac{(N-t)!}{(N+f)!}.$$

Since

$$\frac{(N-t)!}{(N+f)!} = \prod_{h=-t+1}^{f}\frac{1}{N+h} = N^{-(f+t)}\exp\left(-\sum_{h=-t+1}^{f}\log(1+h/N)\right),$$

and $\sum |h|/N = O(f^2/N)$, this becomes

$$2^{s}N^{2t-(f+t)}e^{O(f^2/N)} = (2/N)^{s}e^{O(f^2/N)}.$$

If $h = |R \cap C|$, the vertices carrying both u, v number $f - 2s + h$, and those carrying just one factor number $2s - 2h$. Consequently,

$$c_n^{f-2s+h}L_n^{2s-2h} = c_n^{f-2s}L_n^{2s}(c_n/L_n^2)^h \le c_n^{f-2s}L_n^{2s}.$$

Summing over all choices of internal edges gives

$$\begin{aligned}\sum_{s=0}^{f}\binom{f}{s}^2 s!(2/N)^s c_n^{f-2s}L_n^{2s} &\le c_n^f\sum_{s=0}^{f}\frac{1}{s!}\left(\frac{2L_n^2f^2}{c_n^2N}\right)^s \\ &\le c_n^f\exp\left(\frac{2L_n^2f^2}{c_n^2N}\right).\end{aligned}$$

Below, $\#E$ and $|E|$ both denote the cardinality of a finite set E. When $k = 0$, the deleted set avoids the exceptional set, and the proportion is zero. For $1 \le k \le n$, the proportion satisfies

$$\frac{\#\{U : |U| = k, U \cap F \ne \varnothing\}}{\binom{n}{k}} \le \sum_{x\in F}\frac{\binom{n-1}{k-1}}{\binom{n}{k}} = fk/n.$$

After multiplication by the general convolution weight $K\sqrt{n}\,w_k$, the total is $Kf\sum_k kw_k/\sqrt{n} = o(1)$. Set $Q = S_M/(n+1)$ and $v = \mathbf{1}_N$. Then

$$\det\left(I_N + \frac{2}{n}(I_N + A_M)\right) = (1+1/n)^N\det(I_N+Q)\left[1 + \frac{1}{n+1}v^{\mathsf{T}}(I_N+Q)^{-1}v\right].$$

Using $(I+Q)^{-1} = (I-Q)(I-Q^2)^{-1}$ and the vanishing of a skew-symmetric quadratic form,

$$v^{\mathsf{T}}(I+Q)^{-1}v = v^{\mathsf{T}}(I+Q^{\mathsf{T}}Q)^{-1}v \le \|v\|_2^2 = N.$$

Also, $(1+1/n)^N \le e^{N/n} \le e$, and

$$\det(I + S_M/(n+1)) = \prod_j\left(1 + \frac{\lambda_j^2}{(n+1)^2}\right) \le \prod_j\left(1 + \frac{\lambda_j^2}{N^2}\right).$$

The three factors are at most e, $\det(I + S_M/N)$, and 2, respectively. Multiplication by the permanent factor $e^{-1}/2$ leaves $\rho_N(S_M)$. The constants in the exceptional-set estimate satisfy

$$\frac{10}{3}\frac{1}{4}\frac{21}{20} = \frac{7}{8} = \frac{175}{200}, \qquad \frac{1}{25} + \frac{1}{25} = \frac{16}{200}, \qquad \frac{175+16}{200} = \frac{191}{200}.$$

Suppose that, after deleting the exceptional vertices and a short set, the distance of every exceptional vertex's core-neighbor proportion from $[0, 1/20] \cup [19/20, 1]$ is at most ϵ_n, where $\epsilon_n = O((f + k_n)/n)$. On $q \in [0, 1]$, the derivative of $q(1 - q)$ has absolute value at most one. Thus

$$q_x(1 - q_x) \le \frac{19}{400} + \epsilon_n.$$

Using the same nonnegative weight deviation $\beta_n = O(\sqrt{\log n/n})$,

$$\begin{aligned} u_x v_x &= 4(q_x + \beta_n)(1 - q_x + \beta_n) \\ &= 4q_x(1 - q_x) + 4\beta_n + 4\beta_n^2 \\ &\le \frac{19}{100} + 4\epsilon_n + 4\beta_n + 4\beta_n^2 =: c_n. \end{aligned}$$

Moreover, $u_x, v_x \le 2 + 2\beta_n =: L_n$. Since $\epsilon_n, \beta_n \to 0$, choose a uniform threshold such that

$$4\epsilon_n + 4\beta_n + 4\beta_n^2 < \frac{6}{100}, \qquad c_n < \frac{1}{4}.$$

Each term in the additional exponent above also tends to zero:

$$\frac{f^2}{N} = O((\log n)^2/n) \to 0, \quad \frac{2L_n^2 f^2}{c_n^2 N} = O((\log n)^2/n) \to 0.$$

For $\tau \le A \log n, t \le B \log n$, the error terms in Lemma 4.4 are bounded respectively by constant multiples of

$$\frac{\log n + (\log n)^2}{n}, \quad \frac{(\log n)^{3/2}}{\sqrt{n}}, \quad \sqrt{\frac{\log n}{n}}, \quad \frac{\log n + 1}{n}, \quad \frac{(\log n)^{3/2}}{\sqrt{n}}, \quad \frac{(\log n)^2}{n}.$$

The constants are independent of the tournament and deletion sets, so the resulting $o(1)$ error is uniform.

7.4 The score penalty in the remaining class

Proposition 7.3 (Uniform score-sensitive bound). Fix $K_0 > 0$. There are constants K, N depending only on K_0 such that, for $n \ge N$, every tournament with $a = S\mathbf{1}/(n-1)$, $\|a\|_\infty \le 0.9$, and $\tau = \sum_i a_i^2 \le K_0 \log n$ satisfies

$$\frac{H(T)}{\mu_n} \le \rho_n(S) \exp\left(-\frac{\tau}{2} + \frac{K}{n}\right) + B_n,$$

where $B_n \ge 0$ is the graph-independent tail from Lemma 5.2 and $B_n = o(n^{-1})$.

Proof.

The remaining tournaments satisfy

$$\max_i |a_i| \le 0.9, \qquad a = s/(n-1), \qquad \tau = \sum_i a_i^2 \le K \log n.$$

For the fixed K_0 in the statement, all subsequent constants may depend on K_0. The paired product satisfies

$$\Gamma = \prod_i (1 - a_i^2) \le e^{-\tau}.$$

Apply Lemma 4.4 to the whole tournament. The score product is common to every short term. Retaining it outside the sum absorbs errors that, individually, may exceed $1/n$.

For a principal deleted set of size $k \le k_n$, denoted by U, Lemma 4.4 gives

$$\operatorname{per} A[U^c] \le e^{-1} D_n(S) \Gamma \left(\prod_{i \in U} g_i \right) \frac{(n-k)!}{2^{n-k}} \exp(R_\tau + R_{\tau,k}),$$

$$g_i = (1 - a_i^2)^{-1} \le G := 100/19,$$

$$R_\tau \le K[\mathcal{M}_\tau + \sqrt{\tau/n} + 1/n], \qquad R_{\tau,k} \le K[k\sqrt{\tau/n} + (k + k^2)/n].$$

When accumulating errors, we must retain the common factor Γ. Define the nonnegative coefficients

$$c_k = (2/n)^k \sum_{|U|=k} \det(I + A[U]) \prod_{i \in U} g_i.$$

They satisfy $c_0 = 1$ and $c_k \le G^k w_k$. Restoring the falling factorial produces

$$\log \frac{n^k}{(n)_k} = O(k^2/n), \qquad k \le k_n.$$

Set $\delta_n = \sqrt{\tau/n} + 1/n$ and $r_k = R_{\tau,k} + \log(n^k/(n)_k)$. Taking a nonnegative error bound, uniformly for $\tau \le K_0 \log n$ we have $0 \le r_k \le K\delta_n(k + k^2)$ and $\max_{k \le k_n} r_k = o(1)$. Therefore

$$\sum_{k \le k_n} c_k e^{r_k} \le \sum_{k=0}^{n} c_k + K\delta_n \sum_{k \ge 0} G^k w_k (k + k^2) \le \left(\sum_{k=0}^{n} c_k \right) e^{K\delta_n}.$$

The last step uses $\sum_k c_k \ge 1$ and convergence of the corresponding fixed moments. The normalized short path sum is consequently at most

$$\frac{e^{-1} D_n(S) \Gamma}{2} e^{R_\tau + K\delta_n} \det\left(I + \frac{2}{n} \operatorname{diag}(g)(I + A) \right).$$

Summing the error against the finite moments of $G^k w_k$ preserves the stated rate without a logarithmic loss.

We now remove the weights while preserving the score penalty. Set $W = I + (2/n)(I + A)$ and $\Delta = (2/n) \operatorname{diag}(g - 1)(I + A)$. The symmetric part of W is at least I, so $\|W^{-1}\|_{\mathrm{op}} \le 1$. Each row of $I + A$ has Euclidean norm at most $\sqrt{n}$, and $\sum_i (g_i - 1) \le K\tau$. Decomposing Δ as a sum of rank-one row matrices therefore controls its trace norm, the sum of its singular values. Using $\|uv^{\mathsf{T}}\|_* = \|u\|_2 \|v\|_2$, we obtain

$$\|\Delta\|_* \le K\tau/\sqrt{n}.$$

The determinant inequality $|\det(I + E)| \le \exp(\|E\|_*)$ gives

$$\det(W + \Delta) \le \det(W) e^{K\tau/\sqrt{n}}.$$

Both determinants are positive: the symmetric part of W is positive definite, and the weighted determinant has a principal-minor expansion with positive coefficients. The unweighted rank-one bound of Section 7.3, with $N = n$, is $\det(W) \le 2e \det(I + S/n)$.

Combining the factors above and adding only the long-deletion tail at the end yields the uniform score-sensitive upper bound

$$\frac{H(T)}{\mu_n} \le \rho_n(S) \exp\left\{ -\tau + K\left[\frac{1 + \tau + \tau^2}{n} + \frac{\tau^{3/2} + \tau}{\sqrt{n}} + \sqrt{\tau/n} \right] \right\} + B_n. \tag{7.2}$$

For $\tau \le K_0 \log n$ and sufficiently large order, the polynomial error terms other than the constant $1/n$ consume at most $\tau/4$ in total. Young's inequality controls the remaining square-root term:

$$K\sqrt{\tau/n} \le \tau/4 + K^2/n.$$

The total exponent in equation (7.2) is therefore at most $-\tau/2 + K/n$. Applying Proposition 6.6 gives

$$H(T)/\mu_n \le r_n e^{K/n} + B_n = L + O(1/n).$$

This proves Proposition 7.3 and the sharp bound in the remaining class. □

7.5 Completion of the upper bound

The three classes cover all tournaments. High-variance tournaments have normalized path count $o(1/n)$; low-variance tournaments containing exceptional vertices have count strictly below one; and every remaining tournament satisfies the bound $L + O(1/n)$ just proved. Hence

$$P(n) \le \big(L + O(1/n)\big)\mu_n.$$

The threshold in each case is independent of the particular tournament. Taking a common threshold proves the upper bound in Proposition 7.1.

8. The small-score formula and the lower bound

Theorem 8.1 (Small-score path approximation).

There is an absolute constant C with the following property. For every nonnegative function satisfying $\Delta_s(n)/\sqrt{n} \to 0$, denoted by $\Delta_s = \Delta_s(n)$, there is a threshold $n_0(\Delta_s)$ such that every tournament of order $n \ge n_0(\Delta_s)$ with $\|S\mathbf{1}\|_\infty \le \Delta_s(n)$ satisfies

$$\left|\frac{H(T)}{\mu_n} - \rho_n(S)\right| \le C\frac{(\Delta_s(n)+1)^2}{n}.$$

8.1 Proof of the small-score approximation

Proof.

Write $\Delta_s = \Delta_s(n)$. Apply Lemma 4.5 to every principal deleted set $|U| = k \le k_n$. Since $\Delta_s + k_n + 1 = o(\sqrt{n})$, it applies uniformly over all these subsets. After restoring the falling factorial $(n)_k$, the positive path convolution gives

$$\frac{H(T)}{\mu_n} = \frac{e^{-1}D_n(S)}{2}\sum_{k\le k_n}\left(\frac{2}{n}\right)^k \sum_{|U|=k} \det(I + A[U]) + O\left(\frac{(\Delta_s+1)^2}{n}\right).$$

This error rate has no extra logarithmic factor, because the error is at most a fixed constant times

$$\frac{1}{n}\sum_{k\ge 0} w_k(\Delta_s + k + 1)^2 = O\left(\frac{(\Delta_s+1)^2}{n}\right).$$

Restoring the falling factorial contributes $O(n^{-1}\sum k^2 w_k)$, and Section 5.1 bounds the terms with large deleted sets by $o(n^{-1})$. Extending the short generating sum above to all k costs $n^{-2+o(1)}$, because its coefficients are bounded by w_k. Thus

$$\frac{H(T)}{\mu_n} = \frac{e^{-1}D_n(S)}{2}\det\left(I + \frac{2}{n}(I + A)\right) + O\left(\frac{(\Delta_s+1)^2}{n}\right).$$

Set $u = \mathbf{1}/\sqrt{n}$ and $Q = S/(n+1)$. The rank-one determinant lemma gives

$$\det\left(I + \frac{2}{n}(I+A)\right) = (1+1/n)^n \det(I+Q)\left(1 + \frac{n}{n+1}u^{\mathsf{T}}(I+Q)^{-1}u\right).$$

Skew-symmetry gives

$$u^{\mathsf{T}}(I+Q)^{-1}u = u^{\mathsf{T}}(I-Q^2)^{-1}u,$$

The absolute difference of this quantity from one is at most

$$\|Qu\|_2^2 \le \frac{\Delta_s^2}{(n+1)^2}.$$

Also, $(1+1/n)^n = e(1+O(n^{-1}))$. Replacing $S/(n+1)$ by S/n incurs a logarithmic determinant error $O(n^{-1})$, by the bounded sum of the squared normalized frequencies. Consequently,

$$\det\left(I + \frac{2}{n}(I+A)\right) = 2e\det(I+S/n)\left(1+O(n^{-1}+\Delta_s^2/n^2)\right).$$

It remains to justify the uniform constants in the summation and determinant comparison. The following bounds give the required absolute error; only the threshold at which the small-score assumption applies depends on Δ_s.

For every k, using $\Delta_s + k + 1 \le (\Delta_s+1)(k+1)$ gives

$$\frac{1}{n}\sum_k w_k(\Delta_s+k+1)^2 \le \frac{(\Delta_s+1)^2}{n}\sum_k w_k(k+1)^2.$$

The finite moment on the right gives an absolute constant. For the determinant comparison, use $Q^{\mathsf{T}} = -Q$ to write

$$(I+Q)^{-1} = (I-Q)(I-Q^2)^{-1}, \qquad Q(I-Q^2)^{-1} \text{ is skew-symmetric,}$$

$$\begin{aligned} 1 - u^{\mathsf{T}}(I+Q)^{-1}u &= u^{\mathsf{T}}[I - (I+Q^{\mathsf{T}}Q)^{-1}]u \\ &= (Qu)^{\mathsf{T}}(I+QQ^{\mathsf{T}})^{-1}(Qu) \le \|Qu\|_2^2. \end{aligned}$$

Since $Qu = s/[(n+1)\sqrt{n}]$,

$$\|Qu\|_2^2 = \frac{\sum_i s_i^2}{n(n+1)^2} \le \frac{\Delta_s^2}{(n+1)^2}.$$

The determinant error from changing the normalization satisfies

$$\begin{aligned} 0 &\le \log\det(I+S/n) - \log\det(I+S/(n+1)) \\ &= \sum_j \int_{\lambda_j^2/(n+1)^2}^{\lambda_j^2/n^2} \frac{du}{1+u} \\ &\le \sum_j \lambda_j^2\left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right) \\ &= \frac{n(n-1)}{2}\frac{2n+1}{n^2(n+1)^2} = O(n^{-1}). \end{aligned}$$

and

$$0 \le 1 - n\log(1+1/n) = n\int_0^{1/n} \frac{u}{1+u}\,du \le \frac{1}{2n}.$$

The difference between the rank-one factor and two is

$$\left|1 + \frac{n}{n+1}u^{\mathsf{T}}(I+Q)^{-1}u - 2\right| \le \frac{1}{n+1} + \frac{\Delta_s^2}{(n+1)^2}.$$

Multiplying in order and using $D_n(S)\det(I+S/n) < 3$ gives the absolute error in Theorem 8.1. □

8.2 Carousel tournaments

Corollary 8.2 (Carousel lower bound).

For the odd-order carousel tournaments and even-order one-vertex-deletion constructions defined below,

$$H(\mathrm{Car}_n)/\mu_n = L + O(n^{-1}).$$

Proof.

For odd n, take the residue classes modulo n and define the carousel tournament Car_n as follows: for $1 \le j \le (n-1)/2$, direct $i \to i+j$. It is regular, so $\Delta_s = 0$. For even n, let Car_n be obtained by deleting one vertex from Car_{n+1}. Then $\Delta_s = 1$.

Write $n = 2m+1$, with original indices $i = 0, \dots, 2m$. Let $D_{ii} = (-1)^i$. For $i < j$,

$$(DT_n^0 D)_{ij} = (-1)^{i+j}.$$

Use the reordering $p_k \equiv 2k \pmod n$, represented in $0, \dots, n-1$. It gives the sequence $0, 2, \dots, 2m, 1, 3, \dots, 2m-1$. If i, j have the same parity and their positive difference is $2h$, then

$$2^{-1}(j-i) \pmod n = h \in \{1, \dots, m\}.$$

If their parities differ and $i < j$, write $j - i = 2h+1$. Then

$$2^{-1}(j-i) \pmod n = h + m + 1 \in \{m+1, \dots, 2m\}.$$

In the new indices, edges with differences $1, \dots, m$ point forward, and those with differences $m+1, \dots, 2m$ point backward, exactly as in the carousel tournament. A one-vertex deletion selects the corresponding principal submatrix. The restricted D is still a diagonal sign matrix, and the transitive principal submatrix on the remaining ordered indices is still T_n^0. This identifies the spectrum of both parity constructions. Their path counts are then estimated separately by Theorem 8.1.

For either parity, write $S = S_{\mathrm{Car}_n}$. The identity $D_n(S) = \det(I + iS/n)^{-1}$ and the principal-minor polynomial give

$$\begin{aligned}\rho_n(S) &= \frac{\det(I+S/n)}{\det(I+iS/n)} \\ &= \frac{(1+1/n)^n + (1-1/n)^n}{(1+i/n)^n + (1-i/n)^n} = r_n.\end{aligned}$$

For $z \in \{1, -1, i, -i\}$ and $n \ge 2$, the power series for the principal logarithm on $|w| < 1$ gives

$$\begin{aligned}n\log(1+z/n) &= z - \frac{z^2}{2n} + R_{n,z}, \\ |R_{n,z}| &\le \sum_{h=3}^{\infty} \frac{1}{hn^{h-1}} \le \frac{1}{3n^2(1-1/n)} \le \frac{2}{3n^2}.\end{aligned}$$

Exponentiating yields $(1+z/n)^n = e^z(1 - z^2/(2n) + O(n^{-2}))$. Adding the terms in pairs gives

$$(1+1/n)^n + (1-1/n)^n = 2\cosh 1\,[1 - 1/(2n) + O(n^{-2})],$$
$$(1+i/n)^n + (1-i/n)^n = 2\cos 1\,[1 + 1/(2n) + O(n^{-2})].$$

Because $\cos 1 > 0$, the denominator is bounded below by a fixed positive constant for all sufficiently large n. Using

$$\frac{1}{1+u} = 1 - u + \frac{u^2}{1+u},$$

we obtain the more precise spectral ratio

$$r_n = L[1 - 1/n + O(n^{-2})].$$

The term $-L/n$ is the first correction to the spectral ratio. Identifying the corresponding coefficient for the path count requires an error smaller than the $O(n^{-1})$ estimate in Theorem 8.1. Finally,

$$H(\mathrm{Car}_n)/\mu_n = r_n + O(n^{-1}) = L + O(n^{-1}),$$

there is therefore a fixed K_l such that $P(n) \geq (L - K_l/n)\mu_n$. Combining this with the upper bound, take $K = \max(K_l, K_u)$ and $n_0 = \max(N_l, N_u, 2)$, where K_u, N_u and K_l, N_l are the respective constants and thresholds. This proves Theorem 1.1. □

8.3 Degree stability near the maximum

Corollary 8.3 (Degree-variance stability). There are absolute constants C, N such that, for $n \geq N$ and $0 \leq \varepsilon \leq (L-1)/2$, any tournament satisfying $H(T)/\mu_n \geq L - \varepsilon$ has

$$\tau(T) \leq C(\varepsilon + n^{-1}), \qquad V_{\deg}(T) \leq C(\varepsilon n^2 + n).$$

In particular, every tournament maximizing $H(T)$ has $V_{\deg}(T) = O(n)$.

Proof. The assumption gives $H(T)/\mu_n \geq (L+1)/2 > 1$. The first two classes of Section 7 are therefore excluded by a uniform threshold. Proposition 7.3 and Proposition 6.6 give

$$L - \varepsilon - B_n \leq (L + K_r/n)\exp(-\tau/2 + K/n).$$

For sufficiently large n, $B_n \leq 1/n$ and the left-hand side is bounded below by $(L+1)/4$. Taking logarithms and using $\log(1+x) \leq x$ proves $\tau \leq C(\varepsilon + 1/n)$. Multiply by $(n-1)^2/4$ for the variance estimate. After enlarging the threshold so that $K_l/n \leq (L-1)/2$, Corollary 8.2 permits $\varepsilon = K_l/n$ for an exact maximizer. □

For exact maximizers, the conclusion controls mean square degree deviation and gives a maximum single-vertex deviation of order $\sqrt{n}$. Eventual regularity, carousel uniqueness, edit-distance stability, and the $o(\sqrt{n})$ condition in Theorem 8.1 each require further arguments.

8.4 Stability relative to the actual maximum

Put $\delta_0 = (L-1)/4$ and define the normalized deficit

$$\delta_n(T) = \frac{P(n) - H(T)}{\mu_n} \geq 0.$$

Corollary 8.4 (Stability in the extremal window). There are absolute constants C, N such that every tournament of order $n \geq N$ with $\delta_n(T) \leq \delta_0$ satisfies

$$V_{\deg}(T) \leq C\big(\delta_n(T)n^2 + n\big).$$

Consequently, for every fixed $A \geq 0$, tournaments with $P(n) - H(T) \leq A\mu_n/n$ have $V_{\deg}(T) = O_A(n)$ as $n \to \infty$.

Proof. Choose an absolute $K_0 \geq 0$ such that the lower bound in Theorem 1.1 gives $P(n)/\mu_n \geq L - K_0/n$ for all sufficiently large n. Then

$$\frac{H(T)}{\mu_n} \geq L - \varepsilon, \qquad \varepsilon = \delta_n(T) + K_0/n.$$

Increase the common threshold until $K_0/n \leq \delta_0$. Thus $0 \leq \varepsilon \leq 2\delta_0 = (L-1)/2$, and Corollary 8.3 gives

$$V_{\deg}(T) \leq C_0\big(\delta_n(T)n^2 + (K_0+1)n\big).$$

Absorb the fixed factor $K_0 + 1$ into C. For the last assertion, $\delta_n(T) \leq A/n \leq \delta_0$ once $n \geq A/\delta_0$, giving the bound $C(A+1)n$. □

9. The Hessian of the original angular function

The preceding sections express the path count through the spectral ratio ρ and compare it across tournaments. Wormald's fixed-base blow-up asymptotics involve a Hessian of an implicitly defined angular function. We now identify this Hessian in terms of the same ratio: the identity $\sqrt{d/\mathcal{H}} = \rho_d(S)$ connects the spectral comparison to the model constants, while the full matrix formula establishes the positivity required by the asymptotic theorem.

Let A be the adjacency matrix of a regular tournament of odd order $d \geq 3$, and put

$$S = 2A + I - J, \quad C = S/d, \quad P_0 = J/d, \quad z = 2/d.$$

Thus $C^T = -C$ and $C\mathbf{1} = 0$. Define

$$B(y) = I - (A + \tfrac{1}{2}I)\,\mathrm{diag}(y), \qquad K(y) = \det B(y).$$

Near $z\mathbf{1}$, eliminate y_d from $K(y) = 0$, writing $y_d = h(y_1, \ldots, y_{d-1})$. The angular function uses the local logarithm with value $\log z$ at the base point:

$$\widetilde{f}(\theta) = \log h(ze^{i\theta_1}, \ldots, ze^{i\theta_{d-1}}).$$

Theorem 9.2 (Original angular Hessian). The elimination and logarithm are analytic near the base point. With $E = (e_1 - e_d, \ldots, e_{d-1} - e_d)$,

$$\mathbf{H} := \nabla^2 \widetilde{f}(0) = E^T Q E \succ 0, \qquad Q = (I + C^2)(I - C^2)^{-1}.$$

In particular, writing $\mathcal{H} = \det \mathbf{H}$,

$$\mathcal{H} = d\frac{\det(I+C^2)}{\det(I-C^2)}, \qquad \sqrt{d/\mathcal{H}} = \frac{\det(I+C)}{\det(I+iC)} = \rho_d(S) \in \mathbb{Q}_{>0}.$$

Proof. Let $u = \mathbf{1}/\sqrt{d}$, and let the columns of V be an orthonormal basis of $u^\perp$. Every nonzero frequency λ of C satisfies

$$\lambda^2 \le \tfrac{1}{2}\|C\|_F^2 = \frac{d-1}{2d} < \frac{1}{2}.$$

Set $P = (2A+I)/d = P_0 + C$ and introduce $y_j = ze^{t_j}$. In the orthonormal basis (u, V) write

$$(u,V)^T[I - P\,\mathrm{diag}(e^t)](u,V) = \begin{pmatrix} a & b \\ c & D \end{pmatrix}.$$

At zero, $a = b = c = 0$ and $D(0) = V^T(I-C)V$ is invertible. Thus the analytic Schur complement

$$\psi(t) = a(t) - b(t)D(t)^{-1}c(t)$$

has the same local zero set as $K(ze^t)$. Put $U = (I-C)^{-1}$. Differentiation at zero gives

$$a_j = -1/d, \quad a_{jk} = -\delta_{jk}/d, \quad b_j = -e_j^T V/\sqrt{d}, \quad c_k = -V^T C e_k/\sqrt{d}.$$

Since $b = c = 0$, only $b_j D^{-1} c_k + b_k D^{-1} c_j$ survives from the second derivative of $bD^{-1}c$. Using $VD(0)^{-1}V^T C = UC$, we obtain

$$\psi_j(0) = -1/d, \qquad \psi_{jk}(0) = -\frac{\delta_{jk} + (UC)_{jk} + (UC)_{kj}}{d} = -Q_{jk}/d,$$

where $I + UC + (UC)^T = (I+C^2)(I-C^2)^{-1}$. Moreover, with $q = \det(I-C) > 0$,

$$K_{y_d}(z\mathbf{1}) = -q/(dz) = -q/2 \neq 0.$$

The analytic implicit function theorem therefore supplies h, with $h(z,\dots,z) = z$. Choose the logarithm locally and set

$$g(t') = \log[h(ze^{t'})/z], \qquad \psi(t', g(t')) = 0.$$

Implicit differentiation yields $g_j(0) = -1$ and

$$g_{jk}(0) = -(Q_{jk} - Q_{jd} - Q_{dk} + Q_{dd}).$$

The substitution $t' = i\theta$ gives $\widetilde{f}(\theta) = \log z + g(i\theta)$ and $\nabla^2 \widetilde{f}(0) = -\nabla^2 g(0) = E^T Q E$, with the sign supplied by $i^2 = -1$.

On each skew block of frequency λ, Q acts by $(1-\lambda^2)/(1+\lambda^2) > 0$, and $Qu = u$. Therefore $Q \succ 0$ and $E^T Q E \succ 0$. The columns of E are not orthonormal: $E^T E = I_{d-1} + J_{d-1}$ has determinant d. Consequently

$$\det(E^T Q E) = d \det_{u^\perp} Q = d \det Q.$$

Finally, $\det(I+C^2) = \det(I+iC)^2$ and $\det(I-C^2) = \det(I+C)^2$. Both unsquared determinants are positive, as follows blockwise from $1-\lambda^2 > 0$ and $1+\lambda^2 > 0$. Odd principal minors of the rational skew matrix C vanish; hence $\det(I+iC)$ is rational, proving the last assertion. □

The calculation identifies the full Hessian in the original angular coordinates. The expanded Schur-complement calculations and the explicit local elimination are retained in Detailed proof companion §9.

10. Model limits and the carousel constants

10.1 Four fixed-base limits

Use the blow-up models $\mathcal{T}(T^*, m)$ and $\mathcal{R}(T^*, m)$ defined in the introduction, now with a fixed regular base of odd order $d \geq 3$. With $N = dm$, recall the cycle and path normalizations

$$\nu_N = (N-1)!/2^N, \qquad \mu_N = N!/2^{N-1}.$$

Corollary 10.2. For fixed d, the two limits for model $\mathcal{T}$ are $\rho_d(S)$, and those for model $\mathcal{R}$ are $e^{1/d}\rho_d(S)$:

$$\frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N},\ \frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N} \longrightarrow \rho_d(S), \qquad \frac{\mathbb{E}C_{\mathcal{R}}}{\nu_N},\ \frac{\mathbb{E}H_{\mathcal{R}}}{\mu_N} \longrightarrow e^{1/d}\rho_d(S).$$

The regular-block limits are taken through odd m.

Proof. Wormald's Theorem 4 gives the first cycle limit as $\sqrt{d/\mathcal{H}}$ when $\mathcal{H} \neq 0$, now established by Theorem 9.2. His Theorem 5 transfers it to paths, and Theorems 1 and 5 supply the regular-block factor $e^{1/d}$. □

For comparison, Divoux–Wang [12, Proposition 5.1] gives the same constants directly. For $W = (J_N - I_N + S \otimes J_m)/2$, put $D = 2W/(N-1)$ and $B = D - J_N/N$. On the constant direction, the block-constant zero-sum space, and the within-block zero-sum space, B acts respectively as

$$0, \qquad B_0 = \left.\frac{mS - I_d}{N-1}\right|_{\mathbf{1}_d^\perp}, \qquad -I/(N-1).$$

Consequently its operator norm stays below one, and the determinant ratio in their counting formula satisfies

$$R_N = \frac{\det(I - B)}{\sqrt{\det(I - B^T B)}} \longrightarrow e\rho_d(S).$$

Rescaling D to W contributes $((N-1)/N)^N \to e^{-1}$. Their auxiliary-vertex construction gives $\mathrm{path}(D) = (1 + o(1))N^2\,\mathrm{cyc}(D)$, and Wormald's comparison supplies $e^{1/d}$ for regular blocks. Detailed proof companion §10.1 records the determinant factors and the auxiliary-vertex normalization. Theorem 9.2 additionally identifies the original angular Hessian itself.

10.2 Closed form and monotonicity

Proposition 10.3. For odd $d \geq 3$, put $r_d = \rho_d(\mathrm{Car}_d)$ and $\mathfrak{r}_d = e^{1/d} r_d$. Then

$$r_d = \frac{(1 + 1/d)^d + (1 - 1/d)^d}{(1 + i/d)^d + (1 - i/d)^d}.$$

Indeed, the carousel sign matrix is signed-permutation similar to the transitive sign matrix T_d. Its determinant polynomial is

$$\det(I + tT_d) = \frac{(1+t)^d + (1-t)^d}{2};$$

taking $t = 1/d$ and i/d proves the formula. These determinant evaluations are from [12, Section 7 and Lemma 9.1]; the elementary calculations are recorded in Technical appendix A.1–A.2.

Corollary 10.5. The sequence $\mathfrak{r}_d$, over odd $d \geq 3$, is strictly increasing and tends to $L = \cosh 1/\cos 1$.

Proof. The positive frequencies of T_d/d are [11, Lemma 4.2]

$$a_{j,d} = d^{-1} \cot \frac{(2j-1)\pi}{2d}, \qquad 1 \leq j \leq (d-1)/2.$$

Expanding the logarithmic determinant ratio and using $2\sum_j a_{j,d}^2 = 1 - 1/d$ gives

$$\log \mathfrak{r}_d = 1 + 2 \sum_{j=1}^{(d-1)/2} \sum_{r\geq 1} \frac{a_{j,d}^{4r+2}}{2r+1}.$$

For fixed j, $a_{j,d}$ increases with d: writing $c_j = (2j-1)\pi/2$ and $x = c_j/d$, it equals $x\cot x/c_j$, and

$$(x\cot x)' = \frac{\sin x\cos x - x}{\sin^2 x} < 0 \quad (0 < x < \pi/2).$$

Passing from d to $d+2$ increases every existing frequency and adds a positive one. The displayed convergent series has positive coefficients, proving strict increase. The closed form gives the limit L. □

The higher-order expansion of $\mathfrak{r}_d$, its coefficients and uniform Taylor remainder bounds are recorded in the original Detailed proof companion §10.4. There the block limit is taken at fixed d before the base order tends to infinity.

11. The unique expected extremum for a fixed base order

We now compare all tournament bases of a fixed odd order $d \geq 3$ in the models introduced in §1, with the same normalizations. For regular bases, §10 reduces the comparison to $\rho_d(S)$, whose equality cases were determined in §6. To pass to all bases, we first show that nonregularity produces an exponential loss in both models. The positive gap between the resulting limits then gives a common threshold because the set of bases at fixed order is finite.

Corollary 11.2. There is a threshold M_d such that, for every $m \geq M_d$, the carousel base uniquely maximizes each of the four expected counts (cycles and paths in $\mathcal{T}$ and $\mathcal{R}$), up to ordinary tournament isomorphism. In $\mathcal{R}$, only odd m are admissible.

The assertion concerns expected counts. The threshold depends on d, but is common to all bases of that order and all four admissible comparisons.

11.1 Nonregular bases have exponentially small normalized expectations

Let d_i be the base outdegrees. Set

$$b_i = d_i + \tfrac{1}{2}, \quad a_i = 2b_i/d = 1 + s_i/d, \quad s_i = 2d_i - (d-1), \quad \kappa = (\prod_i a_i)^{1/d}.$$

Here $0 < a_i < 2$ and $\sum_i a_i = d$. The bound $\log a \leq a - 1 - (a-1)^2/8$ on $(0,2)$ implies

$$\log\kappa \leq -\frac{\sum_i s_i^2}{8d^3}.$$

For a nonregular base, the zero-sum even integers s_i have squared sum at least 8, so $\kappa < 1$ and $\kappa^N \leq e^{-N/d^3}$.

In model $\mathcal{T}$, the expected adjacency matrix W has row sum $r_i = mb_i - 1/2$ for each row of type i. A Hamilton cycle uses distinct unordered pairs, so independence of internal edges gives $\mathbb{E}C_{\mathcal{T}} = C(W) \leq \operatorname{per} W$.

For a matrix with entries in $[0,1]$, rowwise linear optimization shows that its permanent is bounded by that of a zero–one matrix whose row sums are the rounded-up original sums. Indeed, first increase a row to the target integer sum, then maximize its linear permanent functional on the corresponding hypersimplex; a maximum occurs at a zero–one vertex. Apply this successively to all rows. Brégman's inequality consequently gives

$$\operatorname{per} W \le \prod_{i=1}^{d} (R_i!)^{m/R_i}, \qquad R_i = \lceil mb_i - \tfrac{1}{2} \rceil.$$

For fixed d, $R_i = mb_i + O(1)$ and $R_i \ge m/4$ when $m \ge 2$. Stirling's bounds therefore yield

$$\log \frac{\operatorname{per} W}{N!/2^N} \le N \log \kappa + O_d(\log N).$$

Thus $\mathbb{E}C_{\mathcal{T}}/\nu_N \le N^{O_d(1)}\kappa^N \to 0$. In model $\mathcal{R}$, internal edges are not independent; instead every sample has row sum exactly $mb_i - 1/2$. Apply Brégman's inequality to each sample adjacency matrix, then average, obtaining the same bound. Optional explicit constants are retained in the expanded reading companion §§11.4–11.5.

To treat paths, add a vertex with independent fair edges to every old vertex. For every tournament G of order $N \ge 2$,

$$\mathbb{E}[C_{\text{aug}} \mid G] = H(G)/4, \qquad 4\nu_{N+1} = \mu_N.$$

The two closing edges of a path are distinct, which is why the factor is exactly $1/4$. In the augmented expectation matrix for $\mathcal{T}$, old rows have sum mb_i; in every augmented sample of $\mathcal{R}$, they have sum at most $mb_i + 1/2$. The common rounded targets $R_i' = \lceil mb_i + 1/2 \rceil = mb_i + O(1)$, together with target N for the new row, again allow Brégman's inequality. The old rows contribute $N \log \kappa + O_d(\log N)$ after normalization, and the one new row changes only the polynomial factor. Hence

$$\frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N}, \quad \frac{\mathbb{E}H_{\mathcal{R}}}{\mu_N} \le (N+1)^{O_d(1)}\kappa^N \longrightarrow 0.$$

An optional fully explicit normalization of this bound is in the expanded reading companion §11.6. The degree-based bounds above cover every nonregular base.

11.2 Strict comparison and one common threshold

For regular bases, Corollary 10.2 and the equality classification in §6 show that the carousel has strictly larger limiting constants than every nonisomorphic regular base. For regular tournaments, the equality classification yields ordinary isomorphism, which is the equivalence relevant to the blow-up models. Nonregular bases have limit zero by §11.1, whereas the carousel limits r_d and $e^{1/d} r_d$ are positive.

There are finitely many bases at fixed order d. For each noncarousel base and each of the four expectations, strict separation of the limits supplies an eventual strict inequality. Taking the maximum of this finite set of thresholds gives M_d, with odd m used for $\mathcal{R}$. Finally, a base isomorphism merely relabels the blocks and preserves their independent identical distributions and all path and cycle counts. Thus carousel-isomorphic bases give equal expectations, and only those do. This proves Corollary 11.2. □

Author statement

The paper incorporates two independently developed projects. The path framework and its L/C_* bound are in the version-specific archive [14]; the Wormald project and its overlap with [13,16] are described in the introduction. Xiangyu Ye completed the companion Lean formalization of the earlier path bound, whose precise statement is recorded in the earlier fixed repository revision.

The repository snapshot [17] documents Lean endpoints for 30 numbered results of the pre-revision manuscript, including the sharp path bound, original angular Hessian, path convolution, spectral comparison and fixed-base expectation extrema. Its coverage table specifies objects, hypotheses, quantifiers and coverage boundaries. The associated reproduction instructions fix the toolchain and audit procedure.

The fixed manuscript correspondence distinguishes existing endpoint conclusions from revised written proofs. Theorems 1.1 and 1.3 have existing endpoints; Proposition 7.3, Theorem 1.2 / Corollary 8.3, the even-order variance observation and Corollary 8.4 have no new endpoints. The historical build record documents the run of 7 November 2026, not a compilation of these new statements. No new full Lean build is claimed for this revision. AI assistance was used in exposition and parts of proof checking.

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